AGENTIC AI FOR ROBUST AND SOCIALLY AWARE AUTONOMOUS DRIVING

INDIVIDUAL THESIS

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ABSTRACT

Pedestrian crossing intention prediction is critical for autonomous vehicles operating in complex urban environments, yet prediction performance can degrade when pedestrians are partially occluded by parked vehicles, surrounding traffic, or roadside obstacles. Many existing approaches assume sufficiently visible pedestrians and often produce binary decisions without explicitly considering the reliability of incomplete visual evidence. This research addresses that gap by developing an occlusion aware pedestrian intention prediction framework that integrates observation reliability, predictive uncertainty, and adaptive decision making.

The framework was developed using the Pedestrian Intention Estimation (PIE) dataset and represents each pedestrian sequence over a 30 frame temporal window using a 525 dimensional feature representation combining appearance, spatial, motion, pose/visibility, and probabilistic observation-reliability information. A Transformer-based model estimates crossing intention, while Monte Carlo dropout and temperature scaling are used to quantify predictive uncertainty and improve confidence calibration. A learned selective decision policy then combines intent probability, uncertainty, reliability, and motion evidence to commit to crossing, commit to not crossing, or defer the decision when further evidence is required. Integrated Gradients are used to provide evidence based explanations for the selected action.

Evaluation on held out PIE pedestrian sequences achieved an intent prediction accuracy of 87.59%, while the selective decision policy achieved 88.94% committed accuracy. An end-to-end CARLA runtime implementation further demonstrated operation from raw simulated video without requiring ground truth pedestrian bounding boxes or occlusion labels during inference, while revealing practical limitations related to domain shift and tracking fragmentation. Overall, the individual contribution is an adaptive occlusion aware intention prediction pipeline that integrates perception reliability, uncertainty aware prediction, selective decision making, and explainability to support safer autonomous vehicle reasoning under partial pedestrian visibility.

Keywords: Pedestrian Intention Prediction, Autonomous Vehicles, Occlusion-Aware Perception, Uncertainty Estimation, Explainable AI

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AP	Average Precision
AV	Autonomous Vehicle
CARLA	Car Learning to Act
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit
IG	Integrated Gradients
MC	Monte Carlo
MLP	Multi-Layer Perceptron
mAP	Mean Average Precision
PIE	Pedestrian Intention Estimation
RGB	Red, Green, Blue
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
TPR	True Positive Rate
YOLO	You Only Look Once

1 INTRODUCTION

1.1 Overall project context

Autonomous vehicles are expected to operate safely and efficiently in highly dynamic urban environments where road users continuously interact with one another under uncertain and changing conditions. Unlike controlled road settings, urban environments contain pedestrians, cyclists, human driven vehicles, parked vehicles, roadside infrastructure, intersections, and partially observable situations that can influence autonomous driving decisions. In such environments, safe vehicle operation requires more than detecting surrounding objects. An autonomous vehicle must also interpret the intentions of nearby road users, reason about possible future actions, negotiate shared road space, comply with socially acceptable driving behavior, and adapt its decisions as new evidence becomes available.

Pedestrian interaction represents one of the most challenging aspects of urban autonomous driving because pedestrian behavior is often uncertain and can change rapidly. A pedestrian approaching the road may intend to cross, wait, hesitate, or react to other vehicles and environmental conditions. At the same time, the pedestrian may be partially hidden by parked vehicles, moving traffic, roadside objects, or other pedestrians. Therefore, a decision-making system that relies only on complete visual observations or instantaneous detections may not provide sufficient information for safe interaction. Autonomous driving systems require mechanisms that can reason from incomplete evidence, estimate confidence in their predictions, and adjust their actions according to the level of uncertainty present in the environment.

The overall research project addresses these challenges through an integrated agentic decision-making framework for autonomous vehicles operating in complex urban scenarios. Rather than treating perception, interaction, social behavior, and learning as isolated problems, the project investigates how multiple intelligent components can cooperate to support safer and more adaptive decision-making. The complete solution is organized around four complementary research components. The first component focuses on adaptive pedestrian intention prediction under partial occlusion, where pedestrian behavior is inferred from incomplete visual observations while explicitly

considering visibility and prediction uncertainty. The second component investigates multi-agent negotiation for right-of-way in complex intersections, enabling autonomous agents to reason about conflicts and coordinate movement with other road users. The third component focuses on proactive social-compliance modelling using agentic Theory of Mind, allowing the system to reason about the likely behavior and expectations of surrounding agents and to make socially appropriate decisions. The fourth component investigates federated collective learning between autonomous vehicles using agentic knowledge sharing, enabling multiple vehicles to benefit from distributed experience without requiring all learning data to be centrally collected.

Together, these components form a broader autonomous-driving architecture in which perception, reasoning, interaction, and learning support one another. Information about pedestrian intentions can contribute to vehicle-level risk assessment and decision-making, while negotiation mechanisms can resolve situations in which multiple agents compete for road space. Social-compliance reasoning can further influence how an autonomous vehicle selects an acceptable action, and federated knowledge sharing can allow the system to improve from experiences gathered across multiple vehicles. This integrated approach reflects the complexity of real-world autonomous driving, where safe decisions depend not on a single prediction model but on the coordinated interpretation of uncertain observations, behavioral cues, environmental context, and interactions among multiple agents.

Within this overall project, the present thesis focuses specifically on the first component: adaptive pedestrian crossing-intention prediction in occluded urban scenarios. The component is designed to provide reliable behavioral evidence to the wider autonomous-driving framework when pedestrians are only partially visible. The following section introduces the individual component in greater detail and explains its specific research context, technical focus, and role within the complete solution.

1.2 Individual component context

The individual research component investigated in this thesis is Adaptive Intent Prediction in Occluded Urban Scenarios. Its primary focus is the prediction of pedestrian crossing intention when the pedestrian is only partially visible to an autonomous vehicle. This problem is particularly relevant in dense urban environments, where parked vehicles, moving traffic, roadside objects, and other road users can temporarily obscure important visual cues. In such situations, an autonomous vehicle must make safety critical decisions despite having incomplete and potentially unreliable observations of pedestrian behavior.

The component is designed to estimate whether an observed pedestrian is likely to cross the road while explicitly accounting for the quality of the available evidence. Rather than treating every visual observation as equally reliable, the approach considers pedestrian visibility and occlusion as part of the reasoning process. Temporal behavior is also important because crossing intention cannot always be inferred from a single frame. The component therefore examines pedestrian behavior over a sequence of observations and combines appearance, spatial, motion, pose related, and visibility information to capture changes in behavior over time.

The research was developed primarily using the Pedestrian Intention Estimation (PIE) dataset, which provides naturalistic urban driving scenes containing pedestrians and vehicle centric observations. Because the original dataset does not provide the explicit occlusion representation required for this research, the development process incorporated occlusion aware data preparation and visibility-related evidence to enable the investigation of intention prediction under incomplete observations. Pedestrian sequences were represented over a 30 frame temporal window, allowing the model to reason about behavioral progression rather than relying solely on instantaneous observations.

At the prediction stage, a Transformer based temporal model is used to estimate pedestrian crossing intention from the fused sequence representation. The component further extends conventional intention prediction by incorporating uncertainty estimation and observation reliability into the decision process. Instead of forcing a crossing or not crossing decision for every sequence, the system is

designed to support selective decision making, allowing it to defer a decision when the available evidence is insufficient or uncertain. This provides a more adaptive behavior for safety critical scenarios in which premature commitment may be undesirable.

Explainability is also incorporated into the component so that the evidence influencing a prediction or decision can be examined rather than presenting only an unexplained classification result. In addition to dataset based model development and evaluation, the component was integrated into an end to end runtime pipeline and tested in the CARLA autonomous driving simulation environment. This runtime stage was used to examine how the developed approach behaves when processing simulated scene observations rather than relying only on pre processed dataset inputs.

Within the overall research project, this component functions as an early behavioral reasoning module that provides pedestrian intention evidence, uncertainty information, and observation reliability cues to the wider autonomous driving decision framework. Its role is therefore not only to predict whether a pedestrian may cross, but also to communicate how confidently that prediction can be made when visual evidence is incomplete. This enables downstream decision making components to reason more cautiously and adaptively in complex urban interactions.

1.3 Problem statement

Reliable pedestrian crossing intention prediction is essential for autonomous vehicles operating in urban environments because an incorrect interpretation of pedestrian behavior can lead to unsafe or unnecessarily conservative driving decisions. The problem becomes significantly more difficult when a pedestrian is partially occluded by parked vehicles, moving traffic, roadside structures, or other road users. Under such conditions, important behavioral cues such as body posture, movement direction, and changes in position may be only partially observable or temporarily unavailable. Consequently, the autonomous vehicle must infer intention from incomplete and potentially unreliable visual evidence.

A major challenge is that the quality of pedestrian observations is not constant over time. A pedestrian may be clearly visible in one frame, substantially occluded in the next, and become visible again as either the pedestrian or surrounding objects move. Treating all observations as equally reliable can therefore cause the prediction model to rely on weak or misleading evidence. Furthermore, a single instantaneous observation may not provide sufficient information to distinguish between a pedestrian who is preparing to cross, one who is merely standing near the roadside, or one whose behavior is temporarily ambiguous. The problem consequently requires temporal reasoning across multiple observations while accounting for changes in visibility and motion.

A further safety concern arises when an intention prediction model is required to produce a definitive crossing or not crossing decision even when its evidence is uncertain. A high confidence but incorrect prediction can be particularly problematic in an autonomous driving context because downstream planning may act on the prediction as though it were reliable. Therefore, a useful intention prediction component should not only estimate pedestrian crossing likelihood but should also quantify uncertainty and recognize when the available evidence is insufficient to support a confident decision.

The research problem addressed in this thesis is therefore the development of an occlusion aware pedestrian crossing intention prediction approach capable of reasoning from incomplete temporal observations, accounting for the reliability of visible evidence, estimating prediction uncertainty, and adapting its decision

according to the strength of the available evidence. In addition, the resulting decision should be interpretable so that the factors contributing to a crossing, not crossing, or deferred decision can be examined.

Addressing this problem requires a solution that operates beyond conventional binary classification. It must combine temporal pedestrian information with visibility related evidence, produce an uncertainty aware estimate of crossing intention, and allow the system to postpone commitment when observations do not provide sufficient support for a reliable conclusion. This forms the central research problem investigated by the individual component of this thesis.

1.4 Research gap

Existing pedestrian crossing intention prediction research has made substantial progress through temporal modelling, multimodal feature fusion, attention mechanisms, and Transformer based architectures. These approaches can exploit cues such as pedestrian appearance, pose, motion, bounding-box dynamics, and surrounding scene information to estimate future crossing behavior. However, their effectiveness is strongly influenced by the quality and completeness of the visual observations available to the model. Under partial occlusion, important appearance and pose cues may disappear, motion estimates may become less reliable, and only a limited region of the pedestrian may remain visible. Although recent studies have begun to consider occlusion aware perception, incomplete observations remain a difficult condition for reliable intention prediction.

A significant limitation is that prediction and evidence reliability are often treated as separate concerns. Conventional intention prediction models generally return a crossing or not crossing prediction whenever an input sequence is provided, even when the observable evidence is weak or ambiguous. Consequently, a numerically confident prediction does not necessarily indicate that the underlying visual evidence is sufficiently reliable. This is particularly important in autonomous driving, where acting on an overconfident but poorly supported pedestrian intention estimate may directly influence downstream planning and risk assessment.

Another gap concerns the relationship between occlusion and predictive uncertainty. Occlusion does not affect every observation equally, pedestrian visibility may change considerably across a temporal sequence. Therefore, simply providing a prediction probability is insufficient to represent the reliability of the decision. A more appropriate approach should explicitly model observation quality, quantify predictive uncertainty, calibrate confidence, and use these factors together when deciding whether the currently available evidence is sufficient to support a definitive action.

A further limitation is the tendency to formulate pedestrian intention prediction as a compulsory binary decision. In ambiguous situations, a safety oriented autonomous system should be able to recognize that additional observations are required rather than immediately committing to crossing or non crossing. However, the integration of occlusion aware reliability, calibrated uncertainty, and a learned selective decision mechanism capable of committing or requesting further evidence remains insufficiently addressed as a unified pedestrian intention framework. Similarly, explainability is often considered separately from the core decision process, reducing the ability to determine which observable cues influenced a decision under degraded visibility.

There is also a practical data related challenge. The PIE dataset provides valuable real world pedestrian crossing behavior and temporal driving observations, but it does not provide the explicit occlusion representation required for controlled evaluation across different levels of pedestrian visibility. This limits direct investigation of how intention prediction and confidence change as visual evidence becomes progressively incomplete. An occlusion aware data preparation strategy is therefore required to support systematic experimentation while retaining the realistic pedestrian behavior available in the original dataset.

Accordingly, the research gap addressed in this thesis is the absence of a sufficiently integrated approach that jointly considers partial pedestrian visibility, observation reliability, temporal intention modelling, predictive uncertainty, confidence calibration, selective commit or defer decision making, and explainability. This research addresses that gap by treating pedestrian intention prediction under occlusion not only as the task of determining whether a pedestrian

is likely to cross, but also as the task of determining whether the currently available evidence is reliable enough to justify that decision.

1.5 Aim and objectives

Research Aim

The aim of this research is to develop and evaluate an occlusion-aware pedestrian crossing-intention prediction framework for autonomous vehicles that can reason from incomplete temporal observations, account for observation reliability, quantify predictive uncertainty, adapt its decision according to the strength of available evidence, and provide interpretable explanations for its predictions and decisions.

Research Objectives

To achieve the above aim, the following objectives were defined:

1. To develop an occlusion aware pedestrian sequence preparation process using the PIE dataset, including controlled occlusion modelling and visibility related information to represent varying levels of pedestrian observability.
2. To construct a multimodal temporal representation of pedestrian behavior by combining appearance, spatial, motion, pose/visibility, and observation-reliability information over a fixed temporal sequence.
3. To develop a Transformer based pedestrian crossing intention prediction model capable of learning temporal behavioral patterns from partially observed pedestrian sequences.
4. To estimate and calibrate predictive uncertainty so that the system can distinguish between confident predictions and cases where incomplete or unreliable observations reduce decision confidence.
5. To develop a selective decision mechanism that combines crossing-intention probability, uncertainty, observation reliability, and motion evidence to determine whether to commit to a crossing decision, commit to a non crossing decision, or defer the decision until additional evidence becomes available.
6. To incorporate explainability into the decision process so that the visual and temporal evidence contributing to the resulting prediction or decision can be examined and interpreted.

7. To evaluate the proposed framework experimentally using held out PIE pedestrian sequences and to assess its end-to-end runtime behavior in the CARLA simulation environment under occluded urban scenarios.

1.6 Research questions or hypotheses

The research was guided by the following questions:

RQ1: How can incomplete pedestrian observations caused by partial occlusion be represented using temporal appearance, spatial, motion, pose/visibility, and observation reliability information for crossing-intention prediction?

RQ2: How effectively can a Transformer based temporal model predict pedestrian crossing intention from partially observed pedestrian sequences?

RQ3: How can predictive uncertainty and confidence calibration be incorporated to reflect the reliability of crossing intention predictions when pedestrian visibility is incomplete or ambiguous?

RQ4: How can crossing intention probability, predictive uncertainty, observation reliability, and motion evidence be combined to support a selective decision policy that can commit to a crossing or not crossing decision, or defer the decision when additional evidence is required?

RQ5: To what extent can explainability techniques identify the temporal and feature-level evidence that contributes to the model's crossing intention predictions and selective decisions?

RQ6: How does the proposed occlusion-aware intention-prediction pipeline behave when deployed end-to-end in a CARLA urban simulation environment using raw simulated observations rather than preprocessed dataset inputs?

These research questions collectively address the progression of the proposed component, from occlusion-aware pedestrian representation and temporal intention modelling to uncertainty aware decision making, explainability, and end-to-end runtime evaluation.

1.7 Scope limitations and assumptions

Scope

This research focuses on pedestrian crossing intention prediction for autonomous vehicles under partial visual occlusion. The study concentrates on determining whether an observed pedestrian is likely to cross the road when the available visual evidence is incomplete or varies in reliability over time.

The primary model development and evaluation were conducted using the Pedestrian Intention Estimation (PIE) dataset. Pedestrian observations were organized into fixed temporal sequences of 30 frames, and the proposed framework combined appearance, spatial, motion, pose/visibility, and observation reliability information to represent pedestrian behavior. As the original PIE dataset does not provide the explicit occlusion information required for the study, occlusion aware data preparation and visibility related representations were incorporated to enable controlled investigation under degraded observations.

The research covers the complete individual pipeline from occlusion aware pedestrian representation to crossing intention estimation, predictive uncertainty, confidence calibration, selective decision making, and explainability. The selective decision mechanism allows the system to commit to a crossing decision, commit to a non crossing decision, or defer the decision when the available evidence is considered insufficient. In addition to dataset based evaluation, an end-to-end implementation was tested in the CARLA simulation environment to investigate the behaviour of the developed pipeline when operating on raw simulated urban observations.

Limitations

Several limitations should be considered when interpreting the findings of this research. First, the main experimental development is based on the PIE dataset; therefore, the learned behavioral patterns are influenced by the environmental, geographic, and behavioral characteristics represented within that dataset. Performance may vary when the system is exposed to substantially different pedestrian populations, camera configurations, traffic environments, or weather and lighting conditions.

Second, the occlusion aware experiments rely on generated or estimated visibility information because explicit pedestrian occlusion labels are not provided in the form required by the proposed framework. Although this enables controlled experimentation, synthetic or estimated occlusion cannot fully reproduce all forms of real world occlusion, such as complex interactions between multiple moving objects, unusual body visibility patterns, or severe scene clutter.

Third, runtime evaluation in CARLA introduces a domain difference between the real world imagery used for model development and simulated visual observations. Differences in pedestrian appearance, scene rendering, motion patterns, detection quality, and camera characteristics may therefore affect prediction performance. During end-to-end runtime testing, pedestrian detection and tracking errors, including fragmented tracks, may also influence the quality of temporal features supplied to the intention model.

The research does not constitute validation on a physical autonomous vehicle and does not provide safety certification for real world deployment. The system is intended as a research prototype for evaluating occlusion-aware intention reasoning rather than a production ready autonomous driving safety system. In addition, the primary behavioral prediction task is focused on crossing versus non-crossing intention and does not attempt to model the full range of possible pedestrian activities or long term trajectory planning.

Assumptions

The study assumes that a pedestrian can be detected and associated across a sufficient number of consecutive observations to form a meaningful temporal sequence. It also assumes that the observable appearance, position, motion, visibility, and related behavioral cues contain sufficient information to support short term crossing intention inference.

It is further assumed that reduced pedestrian visibility can be represented through visibility and reliability related features in a manner that is useful for temporal reasoning. Predictive uncertainty is treated as an important indicator of decision reliability however, uncertainty estimates are not assumed to represent all possible sources of risk in an autonomous driving environment.

Finally, the selective decision mechanism assumes that deferring a decision is preferable to forcing a definitive classification when the available evidence is insufficient. This assumption reflects the safety oriented objective of the study, in which the system is expected to distinguish between a prediction that can be supported by the current evidence and a situation in which additional observations are required before committing to a decision.

1.8 Individual contribution

The individual contribution of this thesis is the design, implementation, and evaluation of an adaptive occlusion aware pedestrian crossing intention prediction component for autonomous vehicles operating under incomplete visual observations. The work focuses specifically on enabling pedestrian intention reasoning when visibility is degraded and when the reliability of the available evidence varies over time.

A key contribution of the research is the development of an occlusion aware data preparation and representation process based on the PIE dataset. The work incorporated visibility and reliability related information together with temporal appearance, spatial, motion, and pose related cues to represent pedestrian behavior across a fixed 30 frame observation window. This provided the basis for investigating crossing intention when pedestrian evidence is partially unavailable or unreliable.

The research further contributed a Transformer based temporal intention prediction model together with mechanisms for predictive uncertainty estimation and confidence calibration. Rather than treating the prediction output as sufficient on its own, the component was extended with a learned selective decision mechanism that considers crossing intention probability, uncertainty, observation reliability, and motion evidence before deciding whether to commit to a crossing decision, commit to a not crossing decision, or defer the decision until further evidence becomes available.

Another individual contribution is the integration of explainability into the decision process through feature attribution analysis, allowing the evidence influencing the model's prediction and selected action to be examined. This

supports a more transparent interpretation of the system's behavior, particularly in cases where visibility is limited or the model elects not to make an immediate decision.

The complete individual component was experimentally evaluated using held out PIE pedestrian sequences and was further implemented as an end-to-end runtime pipeline for testing in the CARLA simulation environment. The CARLA stage extended the evaluation beyond preprocessed dataset inputs by examining the behavior of the system when operating on raw simulated observations, including practical challenges associated with detection, tracking, occlusion, and domain differences.

Within the wider group research project, this individual component contributes pedestrian behavioral evidence, prediction confidence, and observation reliability information that can support higher level autonomous driving decisions. The principal research contribution is therefore not only the prediction of pedestrian crossing intention under occlusion, but the development of a framework that also reasons about how reliable the available evidence is and whether that evidence is sufficient to justify a decision.

1.9 Thesis structure

This thesis is organized into a series of chapters that progressively present the research context, development process, experimental evaluation, and contributions of the proposed occlusion-aware pedestrian crossing-intention prediction framework.

Chapter 1 - Introduction presents the overall research context and the individual research component. It defines the problem statement, research gap, aim and objectives, research questions, scope, limitations and assumptions, and the individual contribution of the study.

Chapter 2 - Literature Review and Related Work reviews the theoretical foundations and existing research related to pedestrian intention prediction, occlusion-aware perception, temporal modelling, uncertainty estimation, selective decision-making, and explainable artificial intelligence. It also examines existing techniques and systems and synthesises the research gap addressed by this thesis.

Chapter 3 - Research Methodology describes the research design, datasets and study materials, experimental procedure, evaluation measures, data-analysis methods, responsible-AI considerations, and reproducibility procedures used throughout the study.

Chapter 4 - Product and Solution Design presents the functional and technical requirements of the proposed component, its position within the complete group solution, the major design alternatives considered, and the detailed architecture of the occlusion-aware intention-prediction framework.

Chapter 5 - Implementation, Integration and Testing explains the development environment, technology stack, component implementation, integration with the wider autonomous-driving system, verification activities, and end-to-end runtime testing, including evaluation within the CARLA simulation environment.

Chapter 6 - Results and Validation presents the experimental results obtained from the PIE-based evaluation together with the technical and runtime validation results. It summarises the performance of the intention prediction model, uncertainty and calibration mechanisms, selective decision policy, explainability component, and integrated runtime pipeline.

Chapter 7 - Discussion interprets the experimental findings, compares them with relevant literature and existing approaches, discusses the research and product contributions, and identifies limitations, threats to validity, and implications of the study.

Chapter 8 - Commercialization and Sustainability considers the potential value of the proposed solution, relevant users and application contexts, scalability, intellectual-property and compliance considerations, and the relationship of the research to sustainable development goals.

Chapter 9 - Conclusion and Future Directions summarises the overall findings of the thesis, evaluates the achievement of the research objectives, highlights the individual research and technical contributions, and identifies opportunities for future development and evaluation.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review was conducted as a structured narrative review to establish the theoretical and technical foundations of pedestrian crossing intention prediction under partial occlusion and to position the proposed research within the existing body of work. A structured narrative approach was selected because the research problem spans several related areas, including pedestrian behavior modelling, temporal and multimodal learning, occlusion aware perception, uncertainty estimation, selective prediction, and explainable artificial intelligence. The objective of the review was therefore not to perform a statistical meta analysis, but to critically examine relevant methods, identify their strengths and limitations, and determine how these research areas collectively relate to intention prediction from incomplete visual evidence.

The review primarily considered peer reviewed studies related to pedestrian behavior and autonomous driving, with particular attention to research published during the recent development of deep learning based intention-prediction methods. Earlier foundational studies were also considered where they introduced concepts directly relevant to the proposed framework, including confidence calibration and selective prediction. The literature was selected based on its relevance to one or more stages of the research problem: representing pedestrian behavior, modelling temporal information, handling incomplete observations, estimating prediction reliability, supporting abstention or deferred decision making, and interpreting model behavior.

The review was organized around six principal themes. The first examines pedestrian crossing-intention prediction, including the progression from manually designed behavioral cues to learned multimodal representations. The second investigates temporal and multimodal modelling, focusing on the integration of appearance, pose, bounding-box dynamics, motion, ego-vehicle information, and environmental context. The third considers occlusion-aware pedestrian perception and intention prediction, particularly methods designed to operate when behavioral cues are partially unavailable. The fourth examines predictive uncertainty and confidence calibration, which are important when probability estimates are used to support safety-related decisions. The fifth considers selective prediction and evidence based decision-

making, where a system may defer a prediction rather than being forced to commit under insufficient evidence. Finally, the review examines explainability and runtime applicability, including techniques that can provide insight into the evidence supporting a model decision and considerations associated with transferring dataset-based models to operational autonomous-driving environments.

Relevant studies were assessed according to their input information, modelling approach, treatment of temporal observations, handling of occlusion or missing evidence, uncertainty-awareness, decision strategy, interpretability, datasets, and reported evaluation methodology. Particular attention was given to whether the proposed techniques explicitly considered the reliability of the available observation, rather than considering only classification performance. This distinction is important for the present research because partial occlusion can reduce the quality of visual evidence even when a conventional classifier continues to produce a high numerical confidence value.

Studies that addressed only general object detection or pedestrian localisation without a meaningful connection to behavioural intention, temporal inference, occlusion handling, or reliability estimation were not treated as central literature. Similarly, preference was given to methods whose technical formulation and experimental evaluation were sufficiently described to permit meaningful comparison. Dataset studies and supporting machine learning techniques were included where they directly informed the methodological choices of the proposed framework.

Review theme	Representative concepts and search terms	Relevance to this research
Pedestrian intention prediction	Pedestrian crossing intention, pedestrian behaviour prediction, crossing prediction, pedestrian action recognition	Establishes the core prediction problem and existing approaches

Temporal and multimodal modelling	Temporal pedestrian modelling, feature fusion, spatio temporal attention, Transformer pedestrian intention, pose and motion fusion	Examines how behavioural evidence is integrated across time and modalities
Occlusion aware perception	Occluded pedestrian intention, partial pedestrian visibility, occlusion-aware prediction, incomplete visual evidence	Investigates prediction when pedestrian information is partially unavailable
Uncertainty and calibration	Predictive uncertainty, Monte Carlo dropout, evidential uncertainty, confidence calibration, temperature scaling	Supports estimation of prediction reliability under degraded observations
Selective decision making	Selective prediction, reject option, abstention learning, defer decision, evidence accumulation	Supports commit-or-defer decisions rather than compulsory classification
Explainability and runtime validation	Explainable AI, feature attribution, Integrated Gradients, autonomous-driving simulation, CARLA pedestrian testing	Supports interpretation and practical validation of the developed component

Table 1: Literature review themes and representative search concepts

The selected literature was subsequently synthesised comparatively to identify the strengths, limitations, and remaining gaps across pedestrian intention prediction, occlusion aware perception, uncertainty estimation, selective decision-making, and explainability. This synthesis provides the basis for the theoretical discussion and research gap analysis presented in the following sections.

2.2 Domain and theoretical foundations

Pedestrian crossing-intention prediction lies at the intersection of computer vision, behavioral modelling, temporal machine learning, uncertainty-aware artificial

intelligence, and autonomous-driving decision support. Unlike pedestrian detection, which aims to determine whether and where a pedestrian is present, intention prediction attempts to infer a future behavioral state from observations that may still be incomplete or ambiguous. For an autonomous vehicle, the problem is therefore not limited to recognising a pedestrian but extends to estimating whether the observed behaviour indicates a likely road-crossing action early enough to support downstream planning.

The theoretical basis of this research is built around six closely related concepts: temporal pedestrian behaviour, multimodal evidence representation, partial observability caused by occlusion, predictive uncertainty and calibration, selective decision-making, and interpretable model reasoning. Together, these concepts provide the foundation for understanding why pedestrian-intention prediction under occlusion requires more than conventional binary classification.

Pedestrian Crossing Intention as Temporal Behavioral Inference

Pedestrian intention is not directly observable in the same manner as physical position or appearance. Instead, it must be inferred from behavioural evidence such as pedestrian movement, body orientation, position relative to the road, changes in speed, and interaction with the surrounding traffic environment. Earlier work investigated directly observable behavioural cues, including pedestrian motion and head orientation, while the introduction of the PIE dataset enabled the study of intention from longer ego vehicle centered pedestrian observations.

A fundamental characteristic of crossing intention is that it evolves over time. A single image may show a pedestrian standing close to the road, but this observation alone does not necessarily distinguish between an individual preparing to cross and one who intends to remain stationary. Consecutive observations provide additional evidence through changes in location, posture, appearance, and motion. Pedestrian-intention prediction can therefore be considered a temporal inference problem, where the interpretation of the current state depends partly on evidence accumulated from preceding observations.

This temporal perspective is particularly important under occlusion. When one frame contains limited visual information, evidence from earlier and later observations may still contribute to an estimate of the pedestrian's likely behaviour. The theoretical implication is that the unit of reasoning should not be treated simply as an isolated frame; rather, a sequence of observations should be considered when estimating crossing intention.

Multimodal and Temporal Evidence Representation

Pedestrian behaviour cannot always be represented adequately by a single information source. Visual appearance may provide information about body orientation and posture, while bounding-box position and size indicate the pedestrian's spatial relationship to the observing vehicle. Motion cues reveal changes in movement over time and pose or contextual information can provide further evidence about behavioural state.

Recent pedestrian intention research has therefore moved increasingly toward multimodal representations. combined visual, motion, and contextual features using feature fusion and spatio temporal attention. Cadena et al. incorporated pose, visual context, and ego vehicle information through a graph-based architecture. Other approaches model pedestrian vehicle relationships or progressively combine bounding-box, pose, and ego vehicle information. These studies support the view that complementary temporal and contextual cues provide a richer representation of pedestrian behaviour than appearance information alone.

Temporal deep learning architectures provide a mechanism for learning relationships among these cues across an observation sequence. In particular, Transformer-based models use self-attention to model dependencies between observations without restricting the model to only immediate temporal neighbors. Within the context of pedestrian intention prediction, such temporal reasoning allows changes in appearance, position, motion, and observation quality to contribute jointly to the estimation of crossing behaviour. The proposed research adopts this theoretical principle by treating intention as the outcome of evolving multimodal evidence rather than a decision derived from a single visual cue.

Occlusion, Partial Observability, and Observation Reliability

A central theoretical challenge in urban pedestrian perception is partial observability. Pedestrians may be obscured by parked vehicles, moving traffic, roadside infrastructure, or other road users. Consequently, the amount and quality of information available to an autonomous vehicle can vary considerably over time. Occlusion may remove appearance information, hide important body regions, reduce the reliability of pose estimation, or interrupt motion and tracking information.

Existing research has begun to address this problem. Li et al. considered visibility variation through adaptive fusion of motion and contextual information, while Liu et al. proposed an occlusion-aware approach that reconstructs missing motion patterns before intention prediction. These approaches illustrate an important distinction: the pedestrian's underlying behavioural intention may remain unchanged even though the quality of the observation of that intention has degraded.

This distinction motivates the concept of observation reliability. A fully visible pedestrian and a severely occluded pedestrian should not necessarily contribute equally reliable evidence to a prediction model. Furthermore, visibility should not always be represented as an absolute state. A perception system may itself be uncertain about whether a pedestrian is unoccluded, partially occluded, or heavily occluded. Preserving a probabilistic representation of visibility allows this ambiguity to be propagated into later reasoning rather than converting it prematurely into a hard label. The proposed framework follows this principle by maintaining probabilistic occlusion evidence and mapping it into observation-reliability information.

The theoretical relationship can therefore be expressed conceptually as:

Visibility condition → Observation reliability → Quality of behavioural evidence → Reliability of intention reasoning

This relationship is fundamental to the research because it separates what the pedestrian may intend to do from how confidently that intention can be inferred from the current observation.

Predictive Uncertainty and Confidence Calibration

classification probability alone should not automatically be interpreted as a reliable measure of confidence. Neural networks can produce probability values that are poorly calibrated, meaning that the numerical confidence assigned to predictions does not necessarily correspond to their empirical correctness. demonstrated this calibration problem in modern neural networks, highlighting the importance of calibration when model probabilities are used to support downstream decisions.

Uncertainty becomes particularly important under occlusion because degraded observations can create ambiguity that is not fully represented by the predicted crossing probability. Recent pedestrian-intention research has explicitly incorporated uncertainty. TrEP combines Transformer-based temporal modelling with evidential learning to jointly estimate intention and uncertainty, while uncertainty aware cross modal Transformer approaches have also been investigated.

From a theoretical perspective, uncertainty-aware prediction seeks to distinguish between the model's estimated behavioural outcome and the reliability of that estimate. Multiple indicators can be used to represent this uncertainty, including variation across repeated predictions, predictive entropy, mutual information, probability variance, and decision-margin information. In the proposed research, Monte Carlo dropout provides repeated stochastic predictions at inference time, while temperature scaling is used to improve probability calibration before the results are passed to the decision stage. The underlying principle is that a high crossing probability should not automatically lead to a committed decision if the accompanying uncertainty or observation reliability indicates weak evidence.

Selective Prediction and Temporal Decision Deferral

Conventional classifiers normally produce a prediction for every input. In safety-related applications, however, forcing a model to commit when evidence is insufficient may be undesirable. Selective prediction introduces a reject or abstention option, allowing a system to make predictions only for observations considered sufficiently reliable. SelectiveNet, proposed by Geifman and El-Yaniv, demonstrated how

prediction and rejection can be learned jointly rather than treating abstention purely as a manually defined post-processing rule.

For temporal pedestrian observations, abstention can be interpreted more appropriately as decision deferral. A pedestrian scene continues to evolve, meaning that additional frames may provide evidence that is currently unavailable. The choice is therefore not necessarily between making a prediction and permanently rejecting the sample. Instead, the system may choose to observe further evidence and reconsider the decision using a later temporal window.

This leads to three conceptual actions:

Commit to Crossing → Commit to Not Crossing → Observe More

The Observe More action does not represent a third pedestrian intention class. It represents temporary withholding of a crossing-intention commitment until stronger evidence becomes available. This distinction is central to the proposed research and links uncertainty estimation with temporal evidence accumulation. The existing framework explicitly defines this action as a temporal deferral rather than an additional intent category.

Selective decision-making consequently introduces a trade-off between coverage and reliability. A system that commits to every observation achieves complete coverage but may include decisions made from highly uncertain evidence. A selective system may reduce the proportion of immediate decisions while attempting to improve the reliability of those on which it does commit. This principle provides the theoretical basis for evaluating both committed accuracy and decision coverage later in this thesis.

Explainability of Evidence Based Decisions

The use of complex temporal models and learned decision policies creates an additional requirement: the ability to examine why a particular decision was produced. In autonomous driving research, an unexplained crossing or non-crossing output provides limited insight into whether the system relied on meaningful evidence or on an unreliable feature.

Explainability techniques address this issue by estimating the contribution of individual input features or groups of features to a model output. In the developed framework, Integrated Gradients is used to analyse the evidence contributing to the selected decision. The decision-state features are organised into interpretable groups representing intent evidence, uncertainty, observation reliability, and pedestrian motion. The resulting attribution can therefore indicate which information supported or opposed a particular action.

Importantly, feature attribution should not be interpreted as proof of causality. It explains how the learned model used the supplied features for a particular decision, but it does not establish that those features causally determine real pedestrian behaviour. The explanation component is therefore used as a mechanism for local evidence interpretation, supporting transparency of the learned decision policy rather than making causal claims. This distinction is also maintained in the implementation underlying this thesis.

Foundation for the Proposed Research

The theoretical concepts discussed above are closely connected. Pedestrian crossing intention is inferred from multimodal evidence accumulated over time; occlusion changes the amount and reliability of this evidence; uncertainty estimation reflects ambiguity in the resulting prediction; calibration improves the interpretation of probability values; selective prediction determines whether the current evidence is sufficient for commitment; and explainability provides insight into the evidence influencing the selected action.

Accordingly, the theoretical foundation of this research can be summarised as:

Temporal behavioural evidence → Occlusion-aware reliability → Intention prediction → Uncertainty and calibration → Selective commitment or deferral → Evidence-based explanation

This perspective distinguishes the research from a conventional pedestrian classifier. The problem is treated not only as estimating what the pedestrian is likely to do, but also as evaluating whether the currently observable evidence is sufficiently reliable to justify acting on that estimate.

2.3 Related research

Pedestrian Crossing Intention Prediction Approaches

Research on pedestrian crossing-intention prediction has progressed from approaches based on individually engineered behavioural cues toward deep temporal and multimodal architectures capable of modelling several interacting sources of evidence. This progression reflects an increasing recognition that crossing intention cannot be inferred reliably from pedestrian appearance alone and that motion, body configuration, spatial relationships, vehicle behaviour, and temporal context can provide complementary information.

Varytimidis et al. represent an earlier stage of this development, investigating visual feature extraction and conventional machine learning approaches for pedestrian action and intention recognition. Their study focused particularly on pedestrian motion and head orientation as behavioural cues for determining whether a pedestrian was likely to cross. Using the JAAD dataset, the study reported an accuracy of 72% for head-orientation estimation and 85% for motion detection. This work demonstrated the value of interpretable behavioural cues; however, the approach depended strongly on the availability and quality of specific observable features and did not provide the richer temporal and multimodal reasoning introduced by later deep-learning approaches.

The introduction of the Pedestrian Intention Estimation (PIE) dataset by Rasouli et al. represented an important development for this research area by providing a large-scale benchmark for pedestrian intention estimation and trajectory prediction from an ego-vehicle perspective. PIE subsequently became a common basis for evaluating temporal pedestrian intention models and supports behavioural and contextual analysis of pedestrian interactions in realistic traffic scenes. The availability of such datasets encouraged a shift from analyzing isolated behavioural cues toward modelling pedestrian behaviour across observation sequences.

Subsequent research increasingly explored the fusion of multiple modalities. Yang et al. proposed an approach based on feature fusion and spatio-temporal attention, combining visual, motion, and contextual evidence to model pedestrian crossing

behaviour over time. Cadena et al. further explored structured behavioural representations through Pedestrian Graph+, incorporating pose-related information with image context and ego-vehicle information through a graph-based modelling approach. Together, these studies demonstrated that complementary behavioural and contextual signals could provide a stronger basis for crossing-intention prediction than reliance on a single visual modality.

More recent work has placed increased emphasis on explicitly modelling the interaction between pedestrians and the ego vehicle. Xu et al. proposed the Pedestrian Vehicle Information Modulation (PVIM) network, arguing that pedestrian intention is strongly influenced by interaction with the surrounding traffic environment. Their method includes a pedestrian-vehicle spatial context module for modelling the spatial dynamics between the pedestrian and the ego vehicle, together with a temporal bilinear attention module designed to reduce temporal redundancy and strengthen temporal correlations. Evaluation on the PIE benchmark reported an accuracy of 0.91, approximately two percentage points above the previous best result reported by the authors.

The progression toward richer multimodal reasoning is also evident in the work of Chen et al., who proposed a progressive multimodal token fusion framework for crossing-intention prediction. Their model uses pedestrian bounding boxes, pose information, and ego-vehicle speed as complementary inputs. Modality-specific processing includes mixture expert feature embedding, spatial-temporal graph convolution for pedestrian pose, multi token temporal aggregation, and progressive multimodal feature fusion. Rather than combining all modalities in a single fusion step, the approach progressively enhances and integrates unimodal, bimodal, and trimodal information, enabling interactions among different behavioural cues to be learned more explicitly. The authors specifically selected pedestrian bounding boxes, skeletal pose, and ego vehicle speed to balance prediction performance and computational complexity.

However, the primary emphasis of these approaches is on obtaining increasingly informative representations for predicting pedestrian intention. Although this progression improves the ability to infer crossing behaviour, the prediction problem

becomes more challenging when some of the required appearance, pose, spatial, or motion cues are degraded or unavailable because of occlusion. In such conditions, an additional question emerges beyond feature fusion and classification: how reliable is the evidence on which the prediction is being made? This issue motivates the occlusion aware research examined in the following subsection.

Study	**Focus and approach**	**Relevance to occlusion / uncertainty / decision-making**	**Limitation relative to the present research**
Varytimidis et al. [1]	Uses pedestrian motion and head orientation with conventional visual feature extraction and machine-learning methods for action and intention recognition.	Establishes interpretable behavioural cues for crossing-intention reasoning.	Relies strongly on directly observable cues and does not address degraded observation reliability or uncertainty-aware decision deferral.
Rasouli et al. – PIE [2]	Introduces a large-scale ego-vehicle-based dataset and models for pedestrian intention estimation and trajectory prediction.	Provides the principal benchmark and behavioural context for temporal pedestrian-intention research.	The dataset was not designed with the explicit probabilistic occlusion representation required for the present study.
Yang et al. [3]	Combines visual, motion, and	Demonstrates the value of temporal	Primary emphasis is predictive feature

	contextual information using feature fusion and spatio-temporal attention.	multimodal evidence for crossing-intention prediction.	fusion rather than explicit observation reliability and commit-or-defer reasoning.
Cadena et al. [4]	Uses graph-based modelling with pedestrian pose, visual context, and ego-vehicle information.	Provides structured modelling of complementary pedestrian and contextual cues.	Does not directly address how prediction reliability should change when these cues become unavailable because of occlusion.
Xu et al. – PVIM [5]	Models pedestrian–vehicle spatial dynamics and temporal relationships using pedestrian-vehicle spatial context and temporal bilinear attention.	Explicitly incorporates interaction between the pedestrian and ego vehicle and reports 0.91 accuracy on the PIE benchmark.	Focuses on improving crossing-intention prediction rather than modelling visibility uncertainty or selectively deferring uncertain decisions.
Chen et al. – Progressive Multimodal Token Fusion [6]	Integrates pedestrian bounding boxes, pose, and ego-vehicle speed using modality-specific feature extraction, temporal	Strengthens temporal and cross-modal representation of pedestrian behaviour.	Relies on informative multimodal observations but does not make observation reliability under partial visibility the central decision variable.

	aggregation, and progressive multimodal fusion.		
Li et al. – MMCF-Net [7]	Combines fine-grained micro-motion cues with contextual features and dynamically changes their contribution according to scene complexity.	Explicitly considers occlusion as part of complex-scene prediction; under occlusion the method reduces the influence of less reliable micro-motion cues and increases contextual weighting.	Handles degraded cues through adaptive weighting but avoids explicit uncertainty modelling and does not introduce a learned commit-or-defer decision policy.
Liu et al. [8]	Uses an occlusion-aware diffusion approach to reconstruct missing motion patterns before pedestrian-intention prediction.	Directly addresses missing pedestrian information caused by occlusion.	Emphasises recovering missing evidence for subsequent prediction rather than jointly reasoning about observation reliability, calibrated confidence, and whether a decision should be deferred.
Zhang et al. – TrEP [9]	Combines Transformer-based temporal modelling with evidential learning for pedestrian intention and	Directly incorporates uncertainty into pedestrian-intention prediction.	Uncertainty estimation alone does not determine whether the available evidence is sufficient for operational commitment or whether additional

	predictive uncertainty.		observation should be requested.
Chen et al. [10]	Applies a cross-modal Transformer with uncertainty-aware multi-task learning for pedestrian crossing-intention prediction.	Demonstrates integration of multimodal temporal reasoning and uncertainty-awareness.	Does not by itself establish a selective temporal policy that jointly considers visibility reliability, uncertainty, motion, and intent evidence before commitment.
Guo et al. [11]	Studies confidence calibration in modern neural networks.	Establishes that neural-network confidence values can be poorly calibrated and motivates post-hoc calibration when probabilities are used for decisions.	Provides a general calibration foundation rather than an occlusion-aware pedestrian-intention or temporal decision framework.
Geifman and El-Yaniv – SelectiveNet [12]	Learns prediction and rejection jointly through an integrated selective-prediction mechanism.	Provides the theoretical basis for abstaining when model evidence is insufficient.	A general selective-classification framework; it does not specifically model evolving pedestrian evidence, occlusion reliability, or temporal re-observation in autonomous-driving scenes.

Table 2: Comparative Analysis of Related Research Relevant to the Proposed Study

Table 2 shows that the literature has progressively addressed important individual aspects of pedestrian-intention reasoning, including multimodal temporal representation, pedestrian vehicle interaction, robustness to occlusion, uncertainty estimation, confidence calibration, and selective prediction. However, these capabilities are largely addressed as separate modelling concerns. Occlusion aware approaches primarily attempt to maintain prediction performance when visual evidence is degraded, while uncertainty aware approaches estimate prediction reliability, and selective-prediction research provides a mechanism for rejecting uncertain outputs. The literature therefore motivates a more integrated perspective in which partial visibility is represented as observation reliability, temporal intention is predicted from multimodal evidence, predictive uncertainty is calibrated, and a learned policy determines whether the available evidence justifies commitment or requires further observation. This combined perspective forms the basis for the solution developed in this thesis.

2.4 Existing systems products and techniques

Pedestrian crossing-intention prediction is generally implemented as a combination of several technical components rather than as a single stand-alone system. Within the reviewed literature, existing solutions can be grouped into benchmark datasets, pedestrian perception and tracking techniques, temporal and multimodal intention-prediction models, occlusion and uncertainty-handling methods, and decision-support or explainability techniques. Each addresses a different stage of the overall problem, but no single reviewed system integrates all of these capabilities into a complete occlusion-aware selective decision pipeline.

Benchmark Datasets for Pedestrian Intention Research

The JAAD and PIE datasets are two important resources used in pedestrian behaviour and intention research. JAAD contains urban traffic video sequences captured from a vehicle-mounted camera and has been used to investigate behavioural cues such as pedestrian motion and head orientation. Varytimidis et al., for example, used JAAD to evaluate pedestrian action and intention-related visual features.

The Pedestrian Intention Estimation (PIE) dataset extended this research direction by providing large-scale ego-vehicle-centred pedestrian trajectories together with behavioural and contextual information for intention estimation and trajectory prediction. Pasted PIE has subsequently been used by several crossing-intention approaches, including pedestrian–vehicle interaction models such as PVIM.

These datasets provide realistic temporal pedestrian observations; however, the reviewed sources do not provide the explicit probabilistic visibility and observation-reliability representation required by the present study. Consequently, additional occlusion-aware processing is required when investigating prediction under systematically degraded visibility.

Pedestrian Detection and Temporal Association

An end-to-end intention-prediction system requires pedestrian observations to be obtained from video before behavioural reasoning can take place. This typically involves pedestrian detection followed by temporal association or tracking. In the runtime implementation developed for this research, a fine-tuned YOLO-based detector was used to identify pedestrians, while BoT-SORT was used to maintain temporal association across frames. A learned three-state occlusion estimator was subsequently used to estimate no, partial, and full occlusion probabilities.

This perception stage is important because temporal intention models depend on the continuity and quality of the pedestrian track. Errors in detection or tracking can therefore propagate into later stages even when the intention model itself performs well. Existing perception techniques provide the necessary observations, but they do not independently determine pedestrian intention or whether the resulting evidence is sufficiently reliable for a safety related decision.

Temporal and Multimodal Prediction Techniques

Several modelling techniques have been applied to pedestrian-intention prediction, including conventional machine learning, recurrent neural networks, graph convolutional networks, attention mechanisms, and Transformers. Earlier approaches relied on engineered behavioural cues, whereas more recent systems fuse information

such as bounding boxes, body pose, pedestrian motion, vehicle dynamics, and scene context.

For example, PVIM models spatial relationships between pedestrians and the ego vehicle and uses temporal bilinear attention to strengthen temporal correlations. Progressive multimodal token fusion similarly combines pedestrian bounding boxes, skeletal pose, and ego-vehicle speed through modality-specific feature extraction, temporal aggregation, and staged fusion.

Transformer based architectures are particularly relevant to the present research because self-attention can model relationships across a temporal observation window. However, temporal modelling alone does not address whether the information provided to the model is trustworthy when a pedestrian becomes partially hidden.

Occlusion, Uncertainty and Calibration Techniques

Existing approaches handle degraded observations in different ways. MMCF-Net adapts the weighting between micro-motion and contextual features according to scene complexity. Under stronger occlusion, the contribution of unreliable micro-motion information is reduced and contextual evidence receives greater weight. The method therefore manages degraded cues adaptively, although the authors describe this as implicit rather than explicit uncertainty modelling.

The reviewed literature also includes an occlusion-aware diffusion model, which addresses missing motion information through reconstruction before intention prediction, and TrEP, which combines Transformer-based prediction with evidential uncertainty modelling. Uncertainty-aware cross-modal Transformer methods have likewise been investigated.

In addition to model-specific uncertainty estimation, confidence calibration is important because raw neural-network probabilities may not accurately represent empirical correctness. Temperature-based calibration provides one means of correcting this mismatch. In the present framework, uncertainty information and calibrated probabilities are therefore treated as decision-support evidence rather than as simple classification outputs. The implemented system uses Monte Carlo dropout for stochastic inference and temperature scaling for probability calibration.

Selective Prediction and Explainability Techniques

A further class of techniques concerns what a system should do when prediction evidence is insufficient. Conventional classifiers are normally forced to return a class for every input, whereas selective prediction introduces an abstention or rejection mechanism. SelectiveNet demonstrates the principle of jointly learning prediction and rejection, providing a foundation for systems that do not need to commit to every observation.

For temporally evolving pedestrian observations, this principle can be adapted from permanent rejection to temporary decision deferral. Rather than declaring a new behavioural class, the system can request further observation and reconsider the pedestrian after additional frames become available.

Explainability techniques provide an additional layer of transparency. In the developed framework, Integrated Gradients is used to measure the contribution of policy-state features to the selected action. Related features are grouped into intent, uncertainty, observation-reliability, and motion evidence, allowing the system to report which factors supported or opposed the decision. These explanations describe learned feature attribution and are not interpreted as causal proof.

2.5 Gap synthesis and conceptual framework

The reviewed literature demonstrates substantial progress in pedestrian crossing-intention prediction through multimodal feature fusion, temporal modelling, pedestrian–vehicle interaction analysis, occlusion-aware processing, uncertainty estimation, confidence calibration, and selective prediction. However, these capabilities are predominantly investigated as separate modelling problems. The central gap identified from the review is therefore not the complete absence of methods for occlusion or uncertainty, but the limited integration of these concepts into a unified decision process for temporally evolving pedestrian observations.

First, recent occlusion-aware approaches improve robustness when visual evidence is incomplete, but they commonly focus on compensating for the missing information. MMCF-Net, for example, dynamically shifts the contribution of micro-motion and contextual features under difficult or occluded conditions, while explicitly describing

its handling of uncertainty as implicit rather than an explicit uncertainty model. The occlusion-aware diffusion approach similarly addresses missing pedestrian motion information through reconstruction before intention prediction. These methods demonstrate that occlusion can be addressed during representation learning, but they do not by themselves determine how trustworthy the resulting observation is for subsequent decision-making.

Second, uncertainty-aware approaches recognise that a prediction probability alone is insufficient for representing model reliability. TrEP incorporates evidential uncertainty into Transformer-based pedestrian-intention prediction, while uncertainty-aware cross-modal learning has also been investigated. Calibration research further demonstrates that neural-network confidence values may not correspond accurately to empirical correctness. Nevertheless, estimating or calibrating uncertainty does not automatically specify whether an autonomous system should accept the current prediction or postpone commitment.

Third, selective prediction provides a mechanism for withholding unreliable classifications. SelectiveNet demonstrates that prediction and rejection can be learned jointly. However, general selective-classification methods do not inherently account for the temporal nature of pedestrian observations. In an evolving traffic scene, rejecting an uncertain observation need not represent a permanent failure to classify; instead, the system may obtain additional evidence from subsequent frames and reassess the pedestrian's intention.

A further gap concerns the connection between observation quality and predictive uncertainty. A pedestrian can transition from clearly visible to partially or severely occluded within the same observation sequence. Therefore, observation reliability is itself temporally varying. The reviewed approaches provide mechanisms for multimodal fusion, occlusion robustness, or uncertainty estimation, but the reviewed body of work does not present these as a single reasoning chain in which probabilistic visibility information is explicitly translated into observation reliability and subsequently considered together with intention probability and uncertainty when deciding whether to commit.

Finally, interpretability and practical runtime operation are often treated separately from the central prediction problem. For safety-oriented behavioural reasoning, it is useful not only to produce a decision but also to identify the evidence that supported or opposed it. The framework developed in this research therefore treats explanation as part of the final decision process, with evidence grouped into intention, uncertainty, observation reliability, and motion-related factors.

Accordingly, the principal gap addressed by this research is the need for an integrated pedestrian-intention framework that jointly considers partial observability, explicit observation reliability, temporal multimodal prediction, predictive uncertainty, confidence calibration, selective decision deferral, and interpretable evidence. Rather than assuming that every observation should result in an immediate binary classification, the proposed research treats the sufficiency of the available evidence as an additional decision problem.

Conceptual Framework

Based on the identified gap, the conceptual framework of this research is organised around a progression from incomplete pedestrian observation to evidence-aware decision-making. Pedestrian observations provide visual, spatial, motion, and visibility-related information. Visibility evidence is interpreted in terms of observation reliability, while multimodal temporal information is used to estimate pedestrian crossing intention. Predictive uncertainty and calibrated confidence are then considered together with observation reliability and motion evidence. These factors determine whether the system should commit to a crossing interpretation, commit to a non-crossing interpretation, or defer the decision until additional evidence becomes available. The selected decision is subsequently accompanied by an evidence-based explanation.

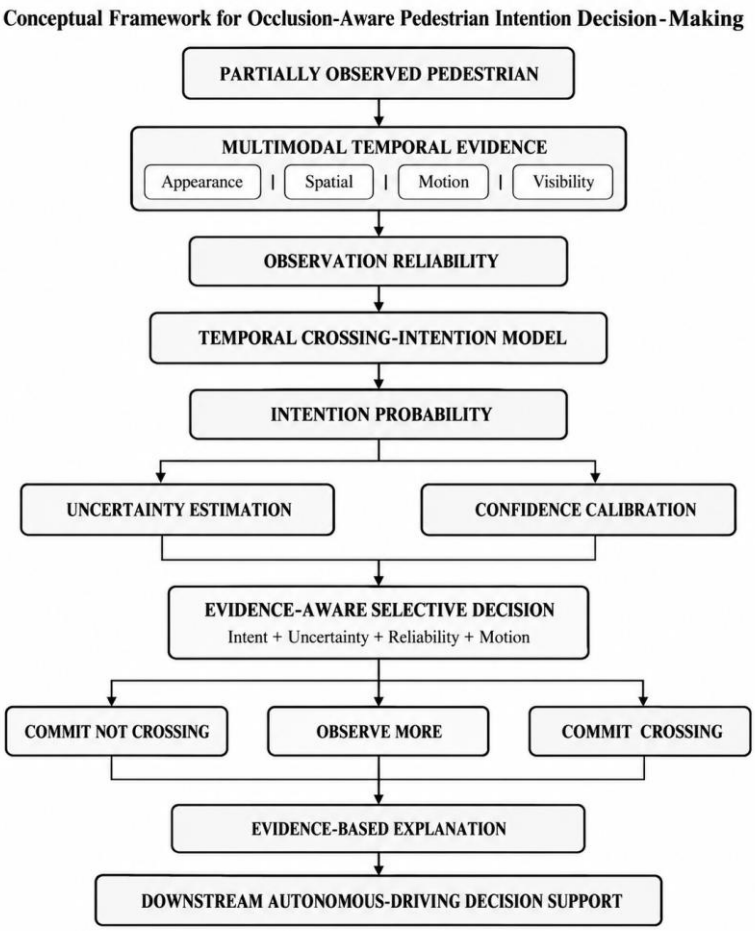


Figure 1: Conceptual Framework for Occlusion-Aware Pedestrian Intention Decision-Making

The conceptual framework deliberately separates pedestrian intention from confidence in the evidence supporting that intention. This distinction enables the system to recognise that a plausible crossing prediction may still be insufficient for immediate commitment when the pedestrian is poorly observed or when predictive uncertainty remains high. Conversely, when the temporal evidence is sufficiently reliable, the system can produce a committed intention signal together with an explanation of the factors that influenced the decision.

This conceptual structure is reflected in the implemented research pipeline, where pedestrian observations are transformed into a 525-dimensional temporal representation, processed by a Transformer-based intent model, evaluated using

uncertainty and calibration mechanisms, and provided to a learned selective policy capable of crossing, non-crossing, or observe more actions.

The resulting framework addresses the identified gap by shifting pedestrian crossing-intention prediction from compulsory classification toward evidence-aware temporal decision-making. The following chapter presents the research methodology used to design, develop, and evaluate this framework.

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopted a predominantly quantitative experimental research design combined with iterative system development and prototype validation. The design was selected because the study aimed not only to investigate pedestrian crossing intention prediction under partial occlusion, but also to design, implement, and experimentally evaluate an adaptive decision making framework capable of reasoning from incomplete and uncertain visual evidence.

The Pedestrian Intention Estimation (PIE) dataset was used as the primary source of pedestrian behavior data. Pedestrian observations corresponding to crossing and non-crossing behavior were retained, while crossing-irrelevant observations were excluded. Continuous pedestrian trajectories were divided into fixed temporal windows of 30 frames so that short-term changes in appearance, position, and motion could be modelled over time. To investigate degraded visibility, controlled synthetic occlusion was introduced during model development, while visibility-related evidence was incorporated into an observation-reliability representation.

Each pedestrian observation was represented using appearance, spatial, motion, and probabilistic observation-reliability information. These features were combined into a temporal input sequence and processed using a Transformer-based crossing-intention model. Predictive uncertainty was estimated through Monte Carlo dropout, while temperature scaling was applied to improve confidence calibration. The calibrated prediction and uncertainty information were then provided to a learned selective decision policy capable of committing to crossing, committing to non-crossing, or requesting additional observation when the evidence was insufficient. Integrated Gradients were subsequently used to provide local evidence-based explanations for the selected decision.

To reduce information leakage between model development and evaluation, the experimental data were separated at the pedestrian identity level, ensuring that pedestrians used for final testing were not included in the corresponding training data.

The framework was evaluated using conventional classification measures together with selective-decision measures such as committed accuracy and coverage.

Finally, a separate end-to-end runtime experiment was conducted using the CARLA simulation environment. During this stage, pedestrian detection, temporal tracking, and occlusion estimation were performed automatically from simulated RGB observations rather than using ground-truth pedestrian information. This runtime evaluation was treated as a practical feasibility test of the complete pipeline and was kept separate from the controlled quantitative evaluation performed on the held-out PIE sequences.

3.2 Data participants or study materials

This research did not involve the direct recruitment of human participants. Instead, it used an existing pedestrian behavior dataset together with synthetically generated occlusion conditions and a simulation environment for runtime validation.

The primary experimental data source was the Pedestrian Intention Estimation (PIE) dataset, which contains pedestrian trajectories recorded in real urban traffic scenes together with behavioral and contextual annotations. For the crossing intention task, observations labelled as crossing and not crossing were retained, while crossing irrelevant observations were excluded. Continuous pedestrian trajectories were segmented into fixed observation windows of 30 consecutive frames to preserve short term temporal changes in pedestrian appearance and movement.

Because the research specifically investigates pedestrian intention under incomplete visibility, synthetic occlusion was introduced during model development to expose pedestrian observations to different levels of visibility degradation. This allowed the effect of incomplete evidence to be studied while retaining the original temporal behavior of the PIE sequences. The final held-out evaluation used pedestrian-level separation to prevent the same pedestrian identities from appearing in both model-development and test data. The held out test set contained 1,152 pedestrian sequences.

For each temporal observation, the study utilized pedestrian appearance, spatial, motion, and observation-reliability information. The resulting representation consisted of a 512-dimensional ResNet-18 appearance descriptor, six spatial features, four

motion features, and three probabilistic observation reliability values. These features formed a 525-dimensional representation for each frame.

A separate set of study materials was used for end-to-end runtime evaluation. The CARLA autonomous-driving simulation environment was used to generate controlled urban driving scenarios in which the complete pipeline could process visual observations under simulated pedestrian and occlusion conditions. Unlike the annotation-based PIE evaluation, the runtime experiment obtained pedestrian detections, tracking information, and occlusion estimates automatically from RGB observations and did not supply ground truth pedestrian identities, bounding boxes, occlusion labels, or crossing labels to the intention pipeline.

Material	Purpose in the Research
PIE Dataset	Primary data source for pedestrian crossing intention model development and quantitative evaluation.
30-frame pedestrian sequences	Temporal representation of short term pedestrian behavior.
Synthetic occlusion	Controlled visibility degradation during model development.
Held out PIE test sequences	Final quantitative evaluation with pedestrian-level separation.
CARLA simulation environment	End-to-end runtime and feasibility evaluation under simulated urban scenarios.

Table 3: Summary of Data and Experimental Materials

3.3 Experimental or study procedure

The experimental procedure was organised as a sequential pipeline covering data preparation, occlusion-aware representation, temporal intention modelling, uncertainty estimation, selective decision-making, explanation, and runtime validation.

The procedure began with the preparation of pedestrian trajectories from the PIE dataset. Observations labelled as crossing and not crossing were retained, while

crossing-irrelevant samples were removed. Continuous pedestrian observations were divided into fixed 30-frame sequences to preserve short-term behavioral changes. Synthetic occlusion was then applied during model development to expose the system to different levels of visibility degradation.

For each pedestrian observation, four main categories of information were extracted: appearance, spatial information, motion, and observation reliability. Appearance features were obtained using a ResNet 18 based representation, while spatial and motion descriptors captured the pedestrian's location, bounding-box characteristics, displacement, speed, and direction. Occlusion evidence was represented probabilistically rather than as a single hard visibility label and was subsequently incorporated into the observation-reliability representation.

The resulting per frame features were combined into the final temporal representation and supplied to the Transformer based intention predictor, which estimated the probability that the pedestrian intended to cross. Data splitting was performed at the pedestrian-identity level so that pedestrians used for final testing were not present in the corresponding model development data.

After training the intention model, prediction uncertainty was estimated using Monte Carlo dropout with 30 stochastic forward passes. Temperature scaling was then applied to improve probability calibration. These calibration parameters were fixed before the final held out evaluation to avoid adapting the model using test-set outcomes.

The calibrated intention predictor was subsequently frozen, and a separate selective decision policy was trained. The policy received intention, uncertainty, observation reliability, and motion information and learned one of three actions: **COMMIT CROSSING, COMMIT NOT CROSSING, or OBSERVE MORE**. Incorrect predictions from the frozen intent model were used to form OBSERVE MORE supervisory targets, whereas correct predictions were assigned the corresponding commit action. The policy-development data were also separated by pedestrian identity.

An explanation stage was then applied to the learned policy using Integrated Gradients. Feature contributions were grouped into intention, uncertainty, observation reliability,

and motion categories so that the evidence supporting or opposing the selected action could be examined.

Finally, the complete pipeline was evaluated in an end-to-end runtime setting. A fine-tuned YOLO-based detector automatically identified pedestrians from RGB observations, while BoT-SORT maintained temporal associations. A learned three class occlusion estimator provided visibility probabilities, and short-term track continuation was used when an already-observed pedestrian temporarily disappeared. The runtime system therefore operated without ground-truth pedestrian identities, bounding boxes, or occlusion labels.

The CARLA simulation stage was treated separately from the controlled PIE evaluation and was used to assess whether the complete perception-to-decision pipeline could operate under simulated urban and occluded conditions rather than to provide an additional dataset level accuracy benchmark.

3.4 Measures and evaluation metrics

The proposed framework was evaluated at several levels because the complete system contains pedestrian perception, crossing-intention prediction, uncertainty-aware selective decision making, and end-to-end runtime operation. Separate measures were therefore used for each stage rather than relying on a single overall accuracy value.

The primary outcome of the study was binary pedestrian crossing intention, represented as crossing or not crossing. The Transformer-based intention model was evaluated using accuracy, precision, recall, F1-score, Receiver Operating Characteristic Area Under the Curve (ROC-AUC), and Average Precision (AP). These measures were selected to provide complementary information about overall correctness, crossing-class detection performance, and discrimination ability in the presence of class imbalance.

Because the final decision policy was allowed to defer uncertain cases through the OBSERVE_MORE action, conventional classification accuracy alone was not sufficient. The selective decision stage was therefore evaluated using committed accuracy, decision coverage, and deferral rate. Coverage represents the proportion of

evaluated observations for which the system commits to either crossing or not crossing:

$$\text{Coverage} = \frac{N_{\text{commit}}}{N_{\text{evaluated}}}$$

The corresponding deferral rate is:

$$\text{Deferral Rate} = 1 - \text{Coverage}$$

Equation 1: Coverage and Deferral Rate Formulation

Committed accuracy measures correctness only among the observations for which the policy makes a committed decision. It must therefore be interpreted together with coverage, since a selective system may improve the reliability of committed predictions by postponing some uncertain cases.

The automatic pedestrian detector was evaluated using precision, recall, mean Average Precision at an IoU threshold of 0.5, and mAP averaged across IoU thresholds from 0.5 to 0.95. The learned three-class occlusion estimator, which classified observations as no, partial, or full occlusion, was evaluated using classification accuracy and macro-F1 score.

Tracking and short-term track continuation were assessed qualitatively during the end-to-end runtime experiments. No quantitative MOTA, IDF1, or HOTA claim was made for the tracking component. Similarly, the complete raw-video runtime was treated as a feasibility evaluation rather than as a separate end-to-end accuracy benchmark.

Component	**Evaluation measures**
Crossing-intention predictor	Accuracy, Precision, Recall, F1-score, ROC-AUC, Average Precision
Selective decision policy	Committed Accuracy, Decision Coverage, Deferral Rate
Pedestrian detector	Precision, Recall

Occlusion estimator	Classification Accuracy, Macro-F1
Tracking / track continuation	Qualitative runtime assessment
End-to-end runtime pipeline	Qualitative feasibility and operational behaviour

Table 4: Evaluation Measures Used in the Study

3.5 Data analysis

The experimental outputs were analysed separately for the crossing-intention model, selective decision policy, automatic perception components, and end-to-end runtime pipeline. Quantitative analyses were performed on the corresponding held-out or validation datasets, while runtime behaviour was examined qualitatively.

For the Transformer-based crossing-intention model, predictions on the held-out pedestrian sequences were compared with the ground-truth crossing and non-crossing labels. Performance was summarised using accuracy, precision, recall, F1-score, ROC-AUC, and Average Precision. The held-out predictions were also examined according to sequence-level occlusion conditions to investigate whether model behaviour changed under different levels of visibility degradation.

The selective decision policy was analysed using the relationship between committed accuracy and coverage. Cases assigned to COMMIT_CROSSING or COMMIT_NOT_CROSSING were treated as committed decisions, whereas OBSERVE_MORE cases were treated as deferred observations. This allowed the analysis to determine whether withholding uncertain cases improved the reliability of the decisions on which the system chose to commit. Committed accuracy was interpreted jointly with coverage rather than as a direct replacement for the base intent accuracy.

The pedestrian detector and occlusion estimator were analysed separately using their respective validation measures. Detector performance was assessed using precision, recall, and mAP-based measures, while the occlusion estimator was analysed using classification accuracy and macro-F1 across the no-, partial-, and full-occlusion states.

Finally, tracking behaviour and complete raw-video operation were analysed qualitatively. The runtime experiments were used to examine whether the complete pipeline could maintain pedestrian observations, propagate occlusion and reliability evidence, generate temporal intent predictions, and produce selective decisions without ground-truth runtime annotations. These observations were treated as feasibility evidence rather than as a separate end-to-end accuracy benchmark.

3.6 Ethics privacy security and responsible AI

This research did not involve the direct recruitment of human participants. Model development was conducted using the existing PIE pedestrian dataset, while the end-to-end runtime experiments were performed in the CARLA simulation environment. The study therefore did not collect new personal information directly from individuals as part of the experimental procedure.

From a privacy perspective, the research focused only on behavioural and visual information necessary for pedestrian crossing-intention prediction. The framework was not designed to identify individual pedestrians or infer unrelated sensitive personal characteristics. Pedestrian identities in the dataset were used only as technical identifiers for maintaining pedestrian-disjoint training and evaluation splits, thereby reducing data leakage rather than supporting person identification.

Appropriate data and model security is also important because the system processes visual data and trained machine-learning artefacts. Experimental data, extracted features, trained models, and evaluation outputs should therefore be stored using controlled access and should not be exposed unnecessarily. The runtime architecture was designed to process pedestrian observations for intention estimation rather than to retain personally identifiable information beyond what is required for temporal analysis.

Several Responsible AI considerations were incorporated into the design. First, the system does not treat every prediction as equally reliable. Prediction uncertainty, observation reliability, and confidence calibration are explicitly considered before the decision policy determines whether to commit to a crossing interpretation or request additional evidence. This reduces the risk of presenting uncertain predictions as

definitive decisions. The OBSERVE_MORE action is therefore an important safety-oriented mechanism, particularly when the pedestrian is heavily occluded or the available evidence is weak.

Second, the framework includes an explainability mechanism based on Integrated Gradients. Explanations identify the local evidence that supports or opposes a selected policy action across intent, uncertainty, observation reliability, and motion related features. These explanations are intended to improve transparency, but they are interpreted as model attributions rather than as causal guarantees.

The study also recognises the possibility of model bias and limited generalisability. Performance obtained from PIE data may not transfer equally to all geographic locations, pedestrian appearances, traffic cultures, weather conditions, camera configurations, or occlusion patterns. Similarly, CARLA provides controlled simulation rather than complete real-world representation. Therefore, the system is not presented as a safety certified autonomous-driving controller.

Finally, the proposed component is intended to provide decision support information to a wider autonomous-driving system rather than independently control vehicle motion. The end-to-end runtime evaluation was treated as a feasibility assessment, and its outputs should not be interpreted as evidence that the system is ready for unsupervised deployment in safety critical real world environments.

3.7 Reproducibility and research artefacts

Reproducibility was considered throughout the development and evaluation of the proposed framework. The experimental workflow was organised into separate stages for data preparation, feature generation, model training, calibration, selective decision learning, and runtime evaluation so that individual stages could be repeated or inspected independently.

To reduce variation caused by data leakage, pedestrian-level dataset splits were maintained so that the pedestrians used for final testing were not included in the corresponding model-development data. The final intent model and calibration settings were frozen before evaluation on the held-out test set, and the selective policy was subsequently trained using outputs from the frozen calibrated predictor.

Research artefacts associated with the component include the PIE sequence-preparation workflow, synthetic-occlusion generation process, extracted temporal feature representations, trained model checkpoints, calibration settings, selective-policy artefacts, evaluation outputs, and runtime test evidence. The wider project also retained held-out split information, frozen model-selection artefacts, scenario seeds, and stored experimental evidence as part of its reproducibility process.

The software implementation was organised modularly so that perception, intention prediction, uncertainty estimation, selective decision-making, and runtime processing could be examined separately. This separation also helps distinguish the quantitative PIE evaluation from the CARLA runtime experiments and prevents runtime feasibility observations from being confused with held out model performance results. The project documentation similarly records independent component modules with clear model and runtime boundaries.

For future reproduction of the experiments, the corresponding source code, dependency information, model checkpoints, dataset-split definitions, configuration values, and CARLA scenario settings should be retained together with the evaluation results. Because some stages involve stochastic processes, such as Monte Carlo dropout and simulation, exact numerical reproduction may depend on the software environment and random-state configuration; however, preserving these artefacts enables the experimental procedure and reported evaluation protocol to be replicated as closely as possible.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements of the proposed component were derived from the central research problem: predicting pedestrian crossing intention when the available visual evidence is incomplete or unreliable. The component was therefore required not only to estimate crossing intention but also to represent observation quality, quantify predictive uncertainty, and determine whether sufficient evidence exists to support a committed decision. The implemented architecture processes temporally evolving pedestrian observations rather than relying on a single fully visible image.

ID	Functional requirement
FR1	The system shall obtain and maintain pedestrian observations across consecutive video frames.
FR2	The system shall estimate pedestrian visibility and represent occlusion using probabilistic evidence rather than relying only on a hard visibility class.
FR3	The system shall construct a temporal pedestrian representation using appearance, spatial, motion, and observation-reliability information.
FR4	The system shall process fixed 30-frame observation sequences to estimate pedestrian crossing intention.
FR5	The system shall estimate predictive uncertainty and calibrate the confidence associated with the crossing intention prediction.
FR6	The system shall determine whether to COMMIT_CROSSING, COMMIT_NOT_CROSSING, or OBSERVE_MORE using intent, uncertainty, reliability, and motion evidence.
FR7	The system shall provide evidence based explanations for the selected decision.
FR8	The runtime pipeline shall operate from raw visual observations without requiring ground truth pedestrian bounding boxes, identities, or occlusion labels as inference inputs.

FR9	The final component shall provide an intention or evidence required signal that can be consumed by downstream autonomous driving components.

Table 5: Functional Requirements of the Proposed Component

These requirements correspond directly to the implemented pipeline, which performs detection and temporal association, visibility estimation, occlusion-aware feature construction, Transformer based intent prediction, uncertainty estimation, and selective decision making. The final implementation additionally supports automatic raw-video operation and explanation generation.

In addition to functional behaviour, several non-functional requirements were considered because the component is intended for safety oriented autonomous driving research.

Requirement	Design expectation
Reliability	The component should account for degraded observation quality instead of treating every visual observation as equally trustworthy.
Uncertainty awareness	Predictions should be accompanied by uncertainty and calibrated confidence information where possible.
Safety-oriented decision behaviour	The system should be able to defer a decision when evidence is insufficient instead of forcing a binary commitment.
Explainability	The factors supporting or opposing the selected decision should be inspectable through feature attribution.
Modularity	Detection, tracking, occlusion estimation, intent prediction, uncertainty estimation, and selective decision stages should remain separable for development and evaluation.

Reproducibility	Dataset splits, frozen model artefacts, configurations, and evaluation evidence should be retained to support repeatable experimentation.
Robustness to partial observation	The system should continue reasoning when pedestrian appearance or motion evidence becomes degraded because of occlusion.
Interoperability	The final action should be represented in a form that can be passed to downstream autonomous driving decision components.

Table 6: Non-Functional Requirements

The resulting requirements define the component as an evidence aware pedestrian intention decision system rather than a conventional binary classifier. In particular, the ability to defer a decision is a core design requirement because uncertain observations can be reconsidered when additional temporal evidence becomes available.

4.2 Component architecture within the complete solution

The proposed component forms the "Understand Intent" pillar of the wider SOCIALMIND DRIVE research framework. The complete project addresses four complementary autonomous-driving challenges: pedestrian intention under occlusion, right-of-way negotiation between autonomous vehicles, socially aware reasoning about road users, and collaborative knowledge sharing between vehicles. Each research component was developed and evaluated independently, while the overall project provides a common autonomous-driving research and demonstration framework.

Project component	Role within the complete solution
Component 1 - Understand Intent	Predicts pedestrian crossing intention from partially occluded and uncertain observations
Component 2 - Negotiate Shared Space	Supports rule-aware multi-agent right-of-way negotiation at complex intersections

Component 3 - Reason Socially	Models pedestrian mental states and socially relevant behavioural cues
Component 4 - Learn Collectively	Supports agentic knowledge sharing and verified learning across autonomous vehicles

Table 7: Position of the Individual Component within the Complete Solution

The complete project architecture positions these four functions around a common agentic autonomous-driving decision-support framework, rather than treating one component as a replacement for the others. Component 1 contributes an early behavioural interpretation of pedestrians, particularly when visibility is incomplete.

Within Component 1, the processing path begins with pedestrian observations obtained from video. Pedestrians are detected and temporally associated, after which visibility evidence is estimated. Appearance, spatial, motion, and observation-reliability information are accumulated over a 30-frame temporal window and processed by a Transformer based intention model. Predictive uncertainty and confidence calibration are then combined with observation reliability and motion evidence in the learned selective decision policy.

The component produces one of three operational outputs:

- COMMIT_CROSSING → CROSSING_INTENT_AVAILABLE
- COMMIT_NOT_CROSSING→ NOT_CROSSING_INTENT_AVAILABLE
- OBSERVE_MORE → MORE_EVIDENCE_REQUIRED

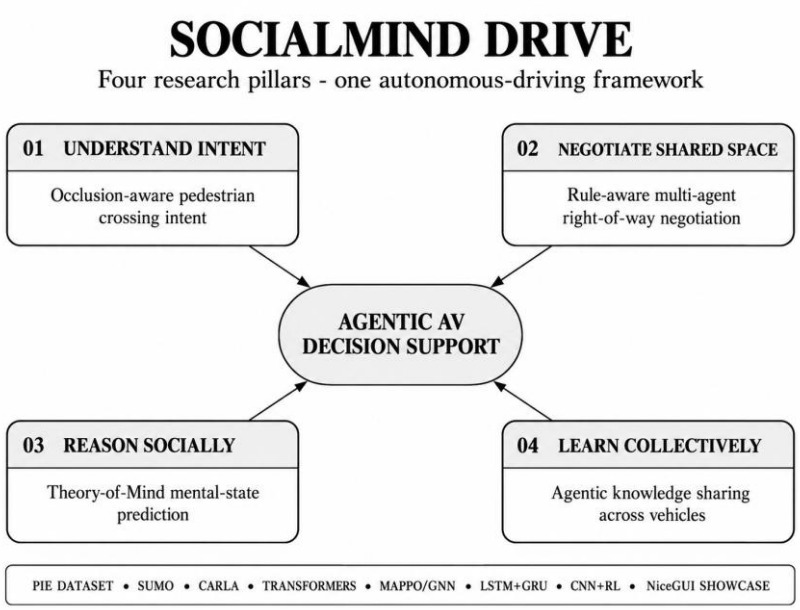


Figure 2: Position of the Adaptive Intent Prediction Component within the SOCIALMIND DRIVE Framework

4.3 Design alternatives and decisions

Several architectural decisions were required when transforming pedestrian crossing-intention prediction from a conventional classification problem into an occlusion-aware decision-support framework. The selected design emphasised temporal evidence, probabilistic observation quality, uncertainty awareness, and the ability to defer decisions when current observations were insufficient.

The final feature representation combines a 512-dimensional ResNet-18 appearance embedding, six spatial features, four motion features, and three Bayesian observation-reliability probabilities, producing a 525-dimensional vector per frame over a 30-frame temporal sequence. This multimodal design was preferred to dependence on visual appearance alone because occlusion can reduce or remove useful appearance information. The framework therefore preserves complementary spatial, motion, and reliability evidence when visual information is incomplete.

A particularly important design decision was to retain probabilistic occlusion information rather than convert the visibility estimator immediately into a fixed hard label. The runtime occlusion estimator produces probabilities for no, partial, and full occlusion, which are incorporated into the observation-reliability stage. This supports

the wider design objective of distinguishing pedestrian intention from the reliability of the evidence used to estimate that intention.

The final decision layer was also deliberately separated from the binary intent predictor. The intent model was first calibrated and frozen, after which a separate selective policy was trained. Incorrect intent predictions in the policy-development data were associated with OBSERVE_MORE, while correct predictions were associated with the corresponding crossing or non-crossing commit action. This design enables the system to treat decision sufficiency as a separate problem rather than assuming that every classifier output should immediately be acted upon.

Overall, the selected design reflects a progression from compulsory classification toward evidence-aware temporal decision-making. The intention predictor answers what the pedestrian is likely to do, while the reliability, uncertainty, and selective-policy stages determine whether the available evidence is sufficient to justify a committed output.

4.4 Detailed design

The proposed component is organised as a sequence of perception, representation, prediction, uncertainty, decision, and explanation stages. During runtime, pedestrians are detected from RGB observations and temporally associated across consecutive frames. A learned occlusion estimator produces probabilistic evidence for no, partial, and full occlusion, while short-term continuation is used when a previously established pedestrian track is temporarily lost.

Each pedestrian observation is represented using four feature groups: a 512-dimensional ResNet-18 appearance descriptor, six spatial features, four motion features, and three Bayesian observation-reliability probabilities. These features are concatenated to form a 525-dimensional representation per frame. Thirty consecutive observations are then combined to produce the temporal input to the Transformer-based intent model.

The Transformer estimates pedestrian crossing intention from this temporal representation. Because the confidence of a prediction may become unreliable under occlusion, the design includes Monte Carlo dropout and temperature scaling. The final

implementation uses multiple stochastic predictions to obtain uncertainty-related evidence, including predictive variance, entropy, mutual information, variation ratio, and decision-margin uncertainty.

The final decision stage is a learned selective policy that combines crossing probability, uncertainty, temporal observation reliability, and pedestrian motion. The deployed policy uses a 15-dimensional state and produces one of three actions: COMMIT_CROSSING, COMMIT_NOT_CROSSING, or OBSERVE_MORE. OBSERVE_MORE represents temporary decision deferral rather than a separate pedestrian-intention class.

Stage	**Main design**
Perception	YOLO-based pedestrian detection and BoT-SORT temporal association
Occlusion	Probabilistic no / partial / full visibility estimation
Reliability	Bayesian observation-reliability representation
Feature construction	512 appearance + 6 spatial + 4 motion + 3 reliability = 525-D
Temporal modelling	30-frame Transformer-based intent prediction
Uncertainty	Monte Carlo dropout and temperature scaling
Decision	Learned 15-D selective policy state
Actions	Commit Crossing / Commit Not Crossing / Observe More
Explanation	Integrated Gradients grouped by evidence type

Table 8: Internal Design of the Proposed Component

4.5 Standards and professional practice

The design and evaluation of the proposed component followed professional practices intended to improve reliability, traceability, reproducibility, and responsible use of AI. The work was developed as a research prototype and decision-support component, and no claim is made that it satisfies a formal automotive safety certification or is ready for direct deployment in a production autonomous vehicle. The project evidence is explicitly limited to stored experimental results and controlled dataset and simulation environments.

A major professional practice was the separation of model development, validation, and final evaluation. Pedestrian-disjoint held-out evaluation was used to reduce identity leakage, while model-selection and calibration artefacts were frozen before final testing. The wider project also retained held-out splits, frozen model-selection artefacts, scenario information, and stored experimental evidence to improve reproducibility and auditability.

The component also follows a modular design practice. Pedestrian detection, tracking, occlusion estimation, observation-reliability modelling, intent prediction, uncertainty estimation, selective decision-making, and explanation are maintained as identifiable stages. This makes it possible to evaluate individual stages independently and prevents the performance of one module from being incorrectly presented as evidence for the entire end-to-end system. The overall project similarly preserves explicit boundaries between its four research components rather than presenting them as a single monolithic algorithm.

Responsible AI principles were reflected in the decision architecture. Rather than treating every model output as reliable, the component incorporates calibrated confidence, predictive uncertainty, and observation reliability. The OBSERVE_MORE action allows a decision to be deferred when the available evidence is insufficient, reducing reliance on compulsory binary predictions under ambiguous observations.

Transparency was supported through the Integrated Gradients explanation stage. Explanations identify the strongest supporting and opposing evidence across intent,

uncertainty, observation reliability, and motion features. These outputs are explicitly treated as local model attributions rather than causal guarantees, which avoids overstating the interpretability provided by the explanation mechanism.

Professional reporting practice was also maintained by bounding research claims to the available evidence. Dataset-based quantitative evaluation, component-level validation, and CARLA runtime feasibility observations were kept distinct. AI-assisted development outputs were subject to verification, and the project documentation explicitly states that experimental results were not fabricated.

Practice	**Application in the research**
Evidence-based reporting	Claims restricted to stored experiments and controlled evaluations
Evaluation separation	Development, validation, and held-out testing kept distinct
Reproducibility	Frozen artefacts, held-out splits, configurations, and experimental evidence retained
Modularity	Perception, prediction, uncertainty, decision, and explanation stages separated
Responsible uncertainty handling	Confidence and uncertainty considered before commitment
Safe deferral	OBSERVE_MORE permits additional evidence to be requested
Transparency	Integrated Gradients used for local feature attribution
Scope limitation	No claim of production safety certification or universal real-world generalisation

Table 9: Professional Practices Applied in the Component

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The proposed component was implemented primarily using a Python-based machine-learning and computer-vision environment. Model development and evaluation were conducted using the PIE dataset, while CARLA was used separately for runtime and simulation-based validation. The implementation was structured so that data preparation, feature extraction, temporal intention modelling, uncertainty estimation, selective decision-making, and end-to-end video processing could be developed and tested as separate modules.

PyTorch was used as the main deep-learning framework for implementing and executing the neural-network components. An ImageNet-pretrained ResNet-18 backbone was used to obtain 512-dimensional pedestrian appearance representations, while the temporal crossing-intention model was implemented using a Transformer architecture. The final representation combined appearance, spatial, motion, and Bayesian observation reliability information into a 525-dimensional feature vector over a 30-frame temporal window.

The computer-vision runtime used a fine-tuned YOLO-based pedestrian detector together with BoT-SORT for temporal pedestrian association. OpenCV-based processing was used within the video and feature-processing workflow, including motion-related computations. A learned three-class occlusion estimator generated probabilities for no, partial, and full occlusion, which were subsequently incorporated into the Bayesian observation-reliability representation.

Uncertainty estimation was implemented using Monte Carlo dropout, with 30 stochastic forward passes during inference, while temperature scaling was used for confidence calibration. The selective decision layer operated on the calibrated intention and uncertainty information, and Integrated Gradients was used to provide evidence-based explanations for the selected policy action.

Development was performed using both local computing resources and a cloud GPU environment. A Vast.ai Ubuntu 22.04 instance with an NVIDIA RTX 3090 GPU was used for computationally demanding CARLA runtime experiments. During the initial

simulation setup, a host using an Intel Xeon E5-2670 v2 processor lacked the AVX2 instruction support required by the selected CARLA runtime and produced an Illegal instruction failure. The runtime was subsequently moved to a host based on an AMD EPYC 7532 processor, where the required CPU instruction support and GPU environment were verified before continuing the experiments. Because the development records contain references to more than one CARLA package version, the thesis should avoid claiming a single exact CARLA version unless it is verified from the final environment log.

Category	**Technology / Resource**	**Purpose**
Programming language	Python	Data preparation, model development, evaluation, and runtime processing
Deep-learning framework	PyTorch	ResNet-18, Transformer and learned decision models
Appearance representation	ResNet-18	512-dimensional pedestrian appearance features
Pedestrian detection	Fine-tuned YOLO-based detector	Automatic pedestrian localisation from RGB frames
Temporal tracking	BoT-SORT	Association of pedestrian observations across frames
Computer vision	OpenCV	Video, image and motion-related processing
Intent prediction	Transformer	30-frame pedestrian crossing-intention modelling
Reliability modelling	Bayesian observation-reliability model	Converts probabilistic occlusion evidence into observation reliability

Uncertainty estimation	Monte Carlo dropout	Predictive uncertainty estimation
Calibration	Temperature scaling	Confidence calibration
Decision layer	Supervised selective policy	Commit crossing, commit not crossing, or observe more
Explainability	Integrated Gradients	Local explanation of policy decisions
Dataset	PIE	Model development and held-out quantitative evaluation
Simulation	CARLA	End-to-end runtime and occlusion-oriented validation
Cloud computation	Vast.ai, Ubuntu 22.04, NVIDIA RTX 3090	GPU-intensive simulation and runtime testing

Table 10: Development Environment and Technology Stack

The development environment therefore supported two complementary forms of evaluation. PIE was used for controlled model development and quantitative evaluation, whereas CARLA was used for external runtime validation of the complete perception-to-decision workflow. The CARLA experiments were not used as a substitute training dataset for the primary intention model.

5.2 Component implementation

The component was implemented as a modular processing pipeline in which pedestrian data preparation, feature construction, temporal intention prediction, uncertainty estimation, selective decision-making, and runtime perception were handled as separate stages. This modular organisation allowed each part of the research pipeline to be developed and evaluated independently before being combined for end-to-end operation.

The implementation began with preprocessing of the PIE dataset. Pedestrian trajectories labelled as crossing and not crossing were retained, while crossing-irrelevant observations were removed. Each trajectory was converted into fixed 30-

frame temporal windows, and synthetic occlusion was introduced during model development to expose the system to different levels of visibility degradation. Pedestrian-level separation was maintained between development and held-out evaluation data.

For each frame, the implementation generated a 525-dimensional pedestrian representation consisting of a 512-dimensional ResNet-18 appearance embedding, six spatial features, four motion features, and three Bayesian observation-reliability probabilities. Thirty consecutive feature vectors were then grouped and supplied to the Transformer-based crossing-intention predictor.

After the intent model was trained, uncertainty estimation was added using Monte Carlo dropout with 30 stochastic forward passes. Temperature scaling was applied after training to calibrate the predicted probabilities. The final frozen implementation used a temperature-scaling parameter of 3.7721 and a calibrated Monte Carlo decision threshold of 0.4751; both values were fixed before evaluation on the final held-out test set.

The selective decision policy was implemented separately after freezing the calibrated intent predictor. Policy-development examples were generated from the frozen model outputs. Incorrect intent predictions were assigned the OBSERVE_MORE target, while correct predictions were assigned either COMMIT_CROSSING or COMMIT_NOT_CROSSING. The policy was therefore trained to distinguish between cases where the current evidence supported commitment and cases where further observation was preferable.

The final decision output was coupled with an Integrated Gradients explanation module. Policy-state features were grouped into intent, uncertainty, observation reliability, and motion evidence. The implementation generated supporting and opposing feature contributions for the selected action and identified the dominant evidence group. These outputs were treated as local model attributions rather than causal explanations.

For end-to-end operation, the annotation-dependent development pipeline was extended with automatic perception. A fine-tuned YOLO-based pedestrian detector

generated pedestrian observations from RGB frames, while BoT-SORT maintained temporal association. A learned three-class occlusion estimator predicted no, partial, and full-occlusion probabilities. When an already-established pedestrian track was temporarily lost, an occlusion-aware continuation mechanism was used to preserve short-term temporal continuity.

5.3 Integration with the other components

The Adaptive Intent Prediction component was developed as one of four complementary research components within the **SOCIALMIND DRIVE** framework. The overall project combines pedestrian-intention reasoning, multi-agent right-of-way negotiation, socially aware behavioural reasoning, and collaborative vehicle learning within a common autonomous-driving research and demonstration environment. However, the scientific algorithms remain independently implemented and evaluated rather than being presented as a single monolithic runtime pipeline

Component 1 exposes its final decision in a form suitable for downstream autonomous-driving reasoning. The selective policy produces one of three operational outcomes: COMMIT_CROSSING, COMMIT_NOT_CROSSING, or OBSERVE_MORE. These are converted respectively into CROSSING_INTENT_AVAILABLE, NOT_CROSSING_INTENT_AVAILABLE, and MORE_EVIDENCE_REQUIRED interface signals. This separation allows the pedestrian-intention module to communicate both a behavioural estimate and whether sufficient evidence exists to justify that estimate.

At the overall system level, this information is conceptually complementary to the other research components. The right-of-way negotiation component addresses vehicle interaction and shared-space decisions, the Theory-of-Mind component addresses socially relevant pedestrian mental states, and the collective-learning component addresses verified knowledge sharing between autonomous vehicles. Component 1 therefore contributes the occlusion-aware pedestrian-intention evidence required by the broader decision-support concept, without claiming that its internal model is directly embedded inside the other component algorithms.

Project-level integration is additionally provided through the SOCIALMIND DRIVE NiceGUI showcase, which offers a common interface for project overview, component access, and evidence-led demonstration. The project architecture deliberately retains separate model and runtime boundaries so that the experimental evidence for each component remains traceable.

For Component 1 specifically, the showcase implementation currently contains a prepared presentation module and a reserved live-integration space. The current implementation records that the Component 1 live runtime is to be connected separately; therefore, this thesis does not claim that the complete Component 1 runtime is already directly launched through the shared showcase. The wider showcase configuration similarly identifies Components 2 and 3 as currently connected for live execution, while the remaining components retain reserved integration spaces.

Integration area	**Current relationship**
Overall SOCIALMIND DRIVE framework	Component 1 contributes occlusion-aware pedestrian-intention reasoning
Downstream decision support	Three explicit commit/defer interface signals are defined
Component 2 – Right-of-way negotiation	Complementary research capability; no verified direct runtime coupling claimed
Component 3 – Theory-of-Mind reasoning	Complementary pedestrian reasoning capability; independently implemented
Component 4 – Collective learning	Complementary fleet-learning capability; independently implemented
Unified NiceGUI showcase	Component 1 presentation space prepared
Component 1 live showcase runtime	Reserved integration space; live connection pending

5.4 Verification and software testing

Verification was performed at both component level and end-to-end runtime level. Individual modules were evaluated independently so that errors in perception, occlusion estimation, intention prediction, and selective decision-making could be distinguished rather than being combined into a single overall result.

The Transformer-based intention model was tested using pedestrian-disjoint held-out PIE sequences, while the selective decision policy was evaluated using committed accuracy and coverage. Automatic perception components were validated separately: the fine-tuned pedestrian detector was assessed using precision, recall, and mAP measures, while the occlusion estimator was assessed using classification accuracy and macro-F1.

Tracking and short-term continuation were verified qualitatively because no separate MOTA, IDF1, or HOTA benchmark was produced. Testing confirmed that an already-established pedestrian track could be maintained for a short period during temporary visual disappearance, although severe occlusion remained one of the main runtime limitations.

End-to-end software testing was then performed directly on raw video. The runtime executed pedestrian detection, tracking, occlusion estimation, observation-reliability modelling, intent prediction, uncertainty estimation, selective decision-making, and explanation without using ground-truth pedestrian boxes, identities, occlusion labels, or crossing labels as runtime inputs.

5.5 Deployment and operation

The developed component was deployed as a **research prototype** rather than as a production autonomous-driving system. Quantitative model evaluation was performed using the PIE dataset, while runtime operation was evaluated separately using simulated and raw-video pedestrian observations. The end-to-end implementation demonstrated that the complete pipeline could operate without supplying ground-truth pedestrian bounding boxes or occlusion labels during inference.

During operation, an RGB frame is first processed by the pedestrian detector and tracker. The occlusion estimator then provides visibility probabilities, which are

transformed into observation-reliability evidence. After a 30-frame pedestrian history has been accumulated, the Transformer produces the crossing-intention estimate. Uncertainty and calibrated confidence are subsequently evaluated by the selective policy before producing a committed decision or requesting further observation.

The operational output is represented through three interface states: CROSSING_INTENT_AVAILABLE, NOT_CROSSING_INTENT_AVAILABLE, and MORE_EVIDENCE_REQUIRED. This allows the component to provide downstream decision-support information without directly controlling vehicle steering, braking, or acceleration.

CARLA was used as the principal simulated runtime environment for evaluating operation under urban and occluded pedestrian scenarios. These runtime experiments were intended to verify the feasibility of the perception-to-decision pipeline and were kept separate from the quantitative held-out PIE evaluation.

At the wider project level, deployment is supported through the **SOCIALMIND DRIVE** demonstration framework. The shared NiceGUI interface provides a common presentation layer for the four research components. However, the current project configuration identifies Component 1 as having a reserved presentation/runtime integration space rather than claiming that its complete live runtime is already launched directly through the common showcase.

The deployed system should therefore be regarded as a simulation-validated research implementation. Further real-world testing, broader scenario coverage, hardware integration, and formal safety validation would be required before considering deployment in a physical autonomous vehicle.

6 RESULTS AND VALIDATION

6.1 Research results

The final Transformer-based pedestrian crossing-intention model was evaluated on 1,152 held-out pedestrian sequences using pedestrian-level separation between model development and testing. The model achieved an overall accuracy of 87.59% and a ROC-AUC of 88.28%, demonstrating that the temporal representation was able to distinguish crossing and non-crossing behaviour under varying observation conditions.

Metric	Result
Accuracy	87.59%
Precision	61.63%
Recall	75.50%
F1-score	67.87%
ROC-AUC	88.28%
Average Precision	69.28%

Table 11: Intent Prediction Performance

The learned selective decision policy was subsequently evaluated using the calibrated intent prediction together with uncertainty, observation-reliability, and motion information. The policy committed to a crossing or non-crossing decision for 87.15% of the observations and achieved 88.94% committed accuracy. The remaining 12.85% were assigned to OBSERVE_MORE, allowing additional temporal evidence to be collected before commitment.

Measure	Result
Base intent accuracy	87.59%
Decision coverage	87.15%
Committed accuracy	88.94%
Deferred observations	12.85%

Table 12: Selective Decision Performance

The committed accuracy should not be interpreted as a direct improvement from 87.59% to 88.94% over the same observations. The base accuracy is calculated over all 1,152 test sequences, whereas committed accuracy is calculated only for the observations on which the selective policy chose to make a final decision. The result therefore represents an accuracy coverage trade off.

Performance was also examined across the available sequence-level occlusion groups. The no/low, partial/medium, and full/high groups achieved base accuracies of 85.31%, 87.04%, and 90.47%, respectively. The pattern was not monotonic with occlusion severity; therefore, the higher accuracy of the full/high group should not be interpreted as evidence that severe occlusion improves prediction. Differences in pedestrian identities, class distributions, motion patterns, and temporal context also influence prediction difficulty.

Overall, the results show that the framework can estimate pedestrian crossing intention from temporally incomplete observations while allowing uncertain cases to be deferred rather than forcing a decision for every sequence.

6.2 Product and technical results

The implemented prototype successfully extended the annotation-based research model into an automatic video-processing pipeline. During runtime, pedestrian detection, temporal association, occlusion estimation, observation-reliability modelling, intent prediction, uncertainty estimation, selective decision-making, and explanation were executed directly from RGB input without supplying ground-truth pedestrian identities, bounding boxes, occlusion labels, or crossing labels.

The automatic perception modules were evaluated separately. The fine-tuned YOLO pedestrian detector achieved approximately 43.3% precision, 38.1% recall, 30.6%. The learned three-class occlusion estimator achieved 58.0% accuracy and approximately 57.8% macro-F1 across the no-, partial-, and full-occlusion classes.

The occlusion estimator results indicate that automatic visibility estimation remains substantially more difficult than the final intention-classification task. Consequently, the estimated visibility output was not treated as an independently reliable hard label. Instead, the probability distribution over the three occlusion states was propagated

through the Bayesian observation-reliability model before being supplied to the temporal predictor.

Tracking was one of the main technical limitations under severe occlusion. Short-term continuation improved temporal coverage when an already-detected pedestrian temporarily disappeared, but the mechanism was not designed to infer pedestrians that had never previously been observed.

In a representative end-to-end occluded sequence, the automatic pipeline processed 270 frames and produced 30 complete intent decisions from automatically established pedestrian tracks. The runtime also maintained a previously observed pedestrian hypothesis during temporary visual disappearance and subsequently processed it using the same 30-frame intention model.

These technical results demonstrate that the complete perception-to-decision architecture is operational as a research prototype. However, the end-to-end runtime observations are treated as feasibility evidence rather than a separate accuracy benchmark, because the quantitative intent, detector, and occlusion-estimator results were obtained from their respective held-out or validation datasets.

6.3 User or stakeholder validation

The primary validation of this research was technical and experimental rather than participant based. The available evidence consists of held-out PIE evaluation, component-level validation, and controlled CARLA runtime experiments. Therefore, no formal user-satisfaction percentage or structured stakeholder usability result is reported in this thesis. The wider project similarly limits its claims to stored experimental evidence and controlled simulation environments.

The main potential stakeholders for the proposed component are autonomous-driving research and development teams, simulation and validation groups, and intelligent-transport research laboratories. For these stakeholders, the practical value of Component 1 is its ability to provide earlier pedestrian-intention evidence under partial occlusion while explicitly communicating uncertainty and the need for further observation when appropriate.

The SOCIALMIND DRIVE showcase was also prepared to provide a common presentation environment through which the overall project and its individual research evidence could be demonstrated. This supports technical review and demonstration, but it should not be interpreted as a formal stakeholder usability study.

Future validation should involve domain experts such as autonomous-driving researchers or intelligent-transport practitioners to evaluate the usefulness, clarity, and trustworthiness of the intent, uncertainty, defer, and explanation outputs in realistic decision-support scenarios.

6.4 Integration results

The individual component was integrated into the wider SOCIALMIND DRIVE research framework at the project and demonstration level. The four research components remain independently implemented and evaluated, while the common NiceGUI showcase provides a unified entry point for accessing component information, demonstrations, and research evidence. This preserves clear model and runtime boundaries rather than presenting the four algorithms as a single end-to-end pipeline.

For Component 1, the latest available showcase implementation contains a dedicated live CARLA demonstration interface. The interface provides controlled crossing and non-crossing scenario selections and connects the showcase to the Component 1 runtime service. The implementation is designed to start the remote CARLA scenario, load the trained model, wait for the runtime stream to become available, and display the resulting live stream within the showcase.

The integration also preserves an important experimental boundary: the selected demonstration scenario controls the CARLA test behaviour, while pedestrian prediction and the final selective-policy action are produced by the trained model rather than being manually predetermined by the interface.

Therefore, the integration work demonstrates a clear path from the standalone Component 1 runtime to the common project demonstration environment. However, the available artefacts primarily verify the implemented integration logic, rather than providing a separate quantitative integration-performance benchmark.

6.5 Summary of findings

The evaluation produced several key findings regarding the effectiveness and practical behaviour of the proposed component. The Transformer-based crossing-intention model achieved 87.59% accuracy and 88.28% ROC-AUC on 1,152 held-out pedestrian sequences. The learned selective decision policy achieved 88.94% committed accuracy at 87.15% coverage, while deferring 12.85% of observations for additional evidence.

The results also showed that performance did not decrease monotonically with increasing sequence-level occlusion. This indicates that prediction difficulty depends not only on visibility severity but also on pedestrian behaviour, class distribution, motion, and temporal context. The learned policy therefore uses intent, uncertainty, observation reliability, and motion jointly rather than relying on occlusion level alone.

At the technical level, the automatic detector and occlusion estimator showed lower performance than the intent model, identifying perception and tracking as important limitations of the complete runtime pipeline. Nevertheless, end-to-end testing demonstrated that the system could process raw video, maintain pedestrian observations through temporary disappearance, and generate complete intent decisions without ground-truth runtime annotations.

The project also demonstrated an integration pathway between Component 1 and the wider SOCIALMIND DRIVE demonstration framework while retaining separate scientific and evaluation boundaries between the four research components.

Overall, the findings support the feasibility of combining occlusion-aware observation reliability, temporal intent prediction, uncertainty estimation, selective decision-making, and explainability within a single pedestrian-intention research framework. However, the results remain bounded to the available PIE evaluation and controlled simulation/runtime experiments and should not be interpreted as evidence of production-ready autonomous-driving safety.

7 DISCUSSION

7.1 Interpretation of findings

The results indicate that the proposed framework can predict pedestrian crossing intention with a useful level of discrimination while also identifying cases in which the available evidence may be insufficient for an immediate decision. The Transformer-based model achieved 87.59% accuracy and 88.28% ROC-AUC, suggesting that the 30-frame temporal representation captured meaningful behavioural information from appearance, spatial, motion, and observation-reliability cues.

The selective decision results are particularly important because they demonstrate a different objective from conventional forced classification. The policy achieved 88.94% committed accuracy at 87.15% coverage, while deferring the remaining 12.85% of observations. This should be interpreted as an accuracy–coverage trade-off, rather than as a direct improvement in overall classification accuracy. The result shows that uncertainty, reliability, and motion information can be used to identify a subset of observations for which commitment is more reliable.

The occlusion-wise analysis also showed that prediction performance did not decrease monotonically as the recorded occlusion category became stronger. In particular, the full/high group achieved a higher base accuracy than the no/low group. This does not imply that severe occlusion makes prediction easier. Instead, it suggests that pedestrian identity, class distribution, motion patterns, temporal context, and visibility interact in determining sequence difficulty.

The comparatively lower results of the automatic detector and occlusion estimator highlight another important finding: the main limitation of end-to-end deployment may arise from the perception layer rather than the intent model itself. The detector and visibility estimator introduce uncertainty before temporal reasoning begins, while strong occlusion can also interrupt track continuity. This supports the decision to retain probabilistic visibility evidence and explicitly represent observation reliability instead of treating perception outputs as perfectly reliable inputs.

Finally, the raw video experiments demonstrate that the complete pipeline is operational beyond annotation-based evaluation. The system was able to perform

detection, tracking, occlusion estimation, reliability modelling, intent prediction, uncertainty estimation, selective decision making, and explanation without ground-truth runtime annotations. However, these experiments demonstrate technical feasibility rather than production level safety or end-to-end accuracy.

Overall, the findings support the central argument of this research: pedestrian crossing-intention prediction under occlusion should be treated not only as a classification problem, but also as a problem of determining whether the available evidence is sufficiently reliable to justify a committed decision.

7.2 Comparison with literature baselines and existing solutions

Previous pedestrian-intention research has progressively moved from individual behavioural cues toward temporal and multimodal representations. Earlier work used observable indicators such as pedestrian motion and head orientation, while later approaches incorporated feature fusion, spatiotemporal attention, graph based reasoning, and pedestrian vehicle interaction modelling. More recent multimodal methods further combine bounding-box, pose, and ego-vehicle information through attention-based fusion mechanisms. These studies demonstrate the value of richer temporal representations, which is consistent with the use of a 30 frame temporal sequence in the proposed framework.

However, the present work differs from conventional intention-prediction approaches by explicitly considering the quality of the observed evidence. Occlusion-aware approaches such as MMCF-Net dynamically balance pedestrian micro-motion and contextual information when visual cues become unreliable, while occlusion aware diffusion methods attempt to reconstruct or reason about information affected by missing observations. These approaches address degraded visibility, but the proposed framework additionally represents occlusion probabilistically and converts this evidence into an observation-reliability representation before intention prediction.

Uncertainty-aware pedestrian intention research has also recognised that prediction probability alone may not provide sufficient information about model confidence. Transformer-based evidential prediction methods such as TrEP explicitly model uncertainty, while cross modal uncertainty-aware approaches incorporate uncertainty

into multimodal pedestrian-intention modelling. The proposed method follows the same broader motivation but uses Monte Carlo dropout together with temperature scaling, allowing uncertainty and calibrated confidence to be supplied to a separate decision stage. Confidence calibration is particularly relevant because modern neural-network probabilities can otherwise be poorly calibrated.

A further distinction is the use of selective decision making. Conventional pedestrian-intention models normally produce a crossing or non-crossing prediction for every observation. Selective prediction research, including SelectiveNet, demonstrates that a model can reject or abstain from predictions when confidence is insufficient. The proposed framework adapts this principle to a temporal pedestrian setting through the OBSERVE_MORE action. Rather than permanently rejecting the observation, the system defers the decision so that the pedestrian can be reconsidered when additional temporal evidence becomes available.

The experimental results showed an overall intent accuracy of 87.59%, while the selective policy achieved 88.94% committed accuracy at 87.15% coverage. These values should not be interpreted as a direct numerical superiority over the cited studies because the compared methods use different datasets, data partitions, input modalities, and evaluation protocols. Instead, the main contribution lies in combining occlusion-aware observation reliability, temporal Transformer modelling, calibrated uncertainty, selective decision deferral, and explanation within one integrated pedestrian-intention framework.

7.3 Research contribution

The main research contribution of this work is an occlusion-aware pedestrian crossing-intention framework that combines prediction with evidence reliability and adaptive decision making. Instead of treating every pedestrian observation as equally informative, the proposed approach explicitly models the quality of the available visual evidence and incorporates this information into the temporal intention prediction process.

A second contribution is the integration of probabilistic occlusion evidence with Bayesian observation reliability. Rather than propagating a hard visibility label directly to the intention model, the framework retains uncertainty in the occlusion estimate and represents observation quality through low, medium, and high reliability probabilities. This provides a more informative representation of partially visible pedestrian observations.

The research also contributes an uncertainty-aware selective decision mechanism. The Transformer-based predictor is combined with Monte Carlo dropout and probability calibration, after which a learned policy determines whether to commit to crossing, commit to non-crossing, or defer the decision through OBSERVE_MORE. The selective policy achieved 88.94% committed accuracy at 87.15% coverage, demonstrating the feasibility of withholding decisions when the available evidence is insufficient.

Another contribution is the inclusion of an explainability layer based on Integrated Gradients. The framework identifies the intent, uncertainty, observation-reliability, and motion evidence that influenced the selected policy action. This extends the system beyond a conventional prediction output by providing insight into why a particular decision was selected.

Finally, the research demonstrates an end-to-end operational pathway from raw visual input to pedestrian intention decision support. Automatic detection, tracking, occlusion estimation, temporal prediction, uncertainty estimation, selective decision-making, and explanation were combined within one runtime architecture. This provides practical evidence that the proposed method can operate beyond annotation-dependent offline evaluation.

Overall, the contribution of this research is not limited to improving pedestrian classification. It reframes pedestrian-intention prediction under occlusion as a problem of predicting behaviour while simultaneously assessing whether the observed evidence is reliable enough to justify acting on that prediction.

7.4 Product contribution

The main product contribution of this work is a functional research prototype for occlusion-aware pedestrian intention analysis that converts the proposed research methods into an operational software pipeline. The prototype accepts raw visual input and performs pedestrian detection, tracking, occlusion estimation, observation-reliability modelling, temporal intention prediction, uncertainty estimation, selective decision-making, and explanation within a single workflow.

A key practical contribution is the use of explicit decision states rather than exposing only a raw crossing probability. The component can produce CROSSING_INTENT_AVAILABLE, NOT_CROSSING_INTENT_AVAILABLE, or MORE_EVIDENCE_REQUIRED. This provides a clearer interface for downstream autonomous-driving decision-support modules because uncertain observations can be distinguished from confident crossing or non-crossing estimates.

The prototype also demonstrates that the trained research models can operate without requiring ground-truth pedestrian bounding boxes, identities, occlusion labels, or crossing labels during runtime. This is important from a product perspective because annotation-dependent evaluation alone would not demonstrate practical deployment feasibility.

Another contribution is the preparation of the component for integration within the wider SOCIALMIND DRIVE demonstration environment. The shared project interface provides a common location for presenting the component, its evidence, and controlled simulation scenarios while keeping its internal research pipeline modular and independently maintainable.

The current product should therefore be regarded as a simulation-oriented decision-support prototype, rather than a production autonomous-driving subsystem. Its value lies in demonstrating how uncertainty-aware pedestrian-intention reasoning can be converted from an experimental model into a modular system that can later be connected to wider perception, planning, or vehicle control architectures.

7.5 Limitations and threats to validity

Several limitations should be considered when interpreting the results of this research. First, the primary quantitative evaluation was conducted using the PIE dataset, and therefore the reported performance is bounded to the characteristics, annotations, and pedestrian behaviours represented in that dataset. Although CARLA and raw-video experiments were used to examine runtime feasibility, they do not provide evidence of real-world deployment performance.

Second, the automatic perception stage remains a significant source of error. The pedestrian detector and learned occlusion estimator achieved lower performance than the final intention model, while severe occlusion could also interrupt pedestrian track continuity. Errors introduced at these early stages may propagate through feature extraction, reliability estimation, and final intention prediction.

The occlusion-wise analysis also presents a validity consideration. The no/low, partial/medium, and full/high groups contain different pedestrians, class distributions, motion patterns, and temporal contexts. Therefore, differences between these groups cannot be attributed to occlusion severity alone. The results should be interpreted as observed behaviour under the available categories rather than as evidence of a direct causal relationship between occlusion level and prediction accuracy.

Another limitation concerns the selective decision policy. Higher committed accuracy is obtained by deferring a portion of the observations, and therefore the result represents an accuracy–coverage trade-off rather than an unconditional improvement over the base predictor. The usefulness of this trade-off may vary depending on how quickly a downstream autonomous-driving system requires a decision.

The end-to-end runtime evaluation was primarily qualitative and feasibility oriented. No separate MOTA, IDF1, or HOTA tracking benchmark was produced, and the raw video tests were not used as a fully annotated end-to-end accuracy benchmark. A larger set of annotated runtime scenarios would be required for stronger system-level validation.

Finally, the current implementation is a research prototype rather than a safety-certified autonomous-driving system. Real-world deployment would require broader

environmental testing, improved perception robustness, latency and hardware evaluation, integration with vehicle planning and control, and formal safety validation.

Overall, these limitations restrict the generalisation of the reported results but do not invalidate the main finding that reliability-aware, uncertainty-aware selective reasoning is feasible for pedestrian crossing-intention prediction under incomplete visual observations.

7.6 Implications

The findings of this research have implications for both pedestrian-intention modelling and autonomous-driving decision support. First, they suggest that prediction confidence alone should not be treated as sufficient evidence for action when pedestrian visibility is degraded. Explicit modelling of observation reliability can provide additional information about whether the current prediction is supported by strong or weak visual evidence.

Second, the results support the use of selective decision-making in partially observable driving scenarios. Allowing the system to request additional evidence rather than forcing an immediate crossing or non-crossing decision provides a practical mechanism for handling ambiguous observations. In a wider autonomous-driving architecture, such a signal could allow planning or control modules to adopt a more cautious strategy until sufficient evidence becomes available.

The research also highlights the importance of strengthening the perception and tracking stages of an end-to-end system. Even when the intention predictor performs well on held-out sequences, errors in pedestrian detection, occlusion estimation, or track continuity can reduce the quality of the evidence supplied to later reasoning stages. Improvements to these front-end components are therefore likely to have a direct effect on overall system reliability.

From a system-design perspective, the modular architecture used in this work allows pedestrian-intention reasoning to remain separate from vehicle control. The component can provide crossing, non-crossing, or additional-evidence signals to downstream modules without directly executing safety-critical vehicle actions. This

supports clearer responsibility boundaries and makes the component easier to evaluate and improve independently.

Overall, the research implies that robust pedestrian-intention prediction under occlusion is better approached as an evidence-aware decision problem rather than as a conventional forced binary classification task. This perspective may also be applicable to other autonomous-driving situations where sensor observations are incomplete, uncertain, or temporarily unavailable.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research investigated pedestrian crossing-intention prediction in urban environments where visual observations may be incomplete because of partial or severe occlusion. The study addressed this problem by developing an occlusion-aware framework that combines pedestrian perception, probabilistic observation reliability, temporal intention prediction, uncertainty estimation, selective decision-making, and explainability.

The proposed approach represented each pedestrian over a 30-frame temporal window using appearance, spatial, motion, and Bayesian observation-reliability information. A Transformer-based model was used to estimate crossing intention, while Monte Carlo dropout and temperature scaling were applied to quantify and calibrate predictive uncertainty. A learned selective decision policy then used intent, uncertainty, reliability, and motion evidence to determine whether to commit to a crossing decision, commit to a not crossing decision, or request additional evidence.

Evaluation on 1,152 held-out PIE pedestrian sequences produced an overall intent-prediction accuracy of 87.59% and a ROC-AUC of 88.28%. The selective policy achieved 88.94% committed accuracy at 87.15% coverage, demonstrating the value of allowing uncertain observations to be deferred rather than forcing a binary prediction in every situation.

The research also extended beyond annotation-based evaluation through an automatic raw-video pipeline incorporating pedestrian detection, tracking, occlusion estimation, temporal inference, uncertainty analysis, selective decision-making, and explanation. These experiments demonstrated the operational feasibility of the proposed architecture, although the perception and tracking stages remain important limitations for future improvement.

Overall, the study demonstrates that pedestrian-intention prediction under occlusion can be strengthened by treating it as an evidence-aware and uncertainty-aware decision problem, rather than solely as a conventional classification task. The resulting framework provides a practical foundation for more cautious and interpretable

pedestrian intention reasoning within future autonomous-driving decision-support systems.

8.2 Achievement of objectives

The objectives defined at the beginning of this research were addressed through the development and evaluation of the proposed occlusion aware pedestrian intention framework.

Objective	Achievement
01 - Prepare pedestrian data suitable for studying intention prediction under occlusion.	Achieved. PIE pedestrian sequences were prepared using 30-frame temporal windows, and controlled occlusion was incorporated to support the investigation of incomplete visual observations.
02 - Develop a temporal representation that incorporates relevant pedestrian and observation-quality information.	Achieved. A 525-dimensional per-frame representation combining 512-dimensional appearance features, spatial information, motion information, and Bayesian observation-reliability probabilities was constructed and processed over 30-frame sequences.
03 - Develop a temporal model for pedestrian crossing-intention prediction.	Achieved. A Transformer-based binary intention predictor was implemented and evaluated on pedestrian-disjoint held-out PIE sequences, achieving 87.59% accuracy and 88.28% ROC-AUC.
04 - Estimate and calibrate predictive uncertainty.	Achieved. Monte Carlo dropout was used to derive multiple uncertainty measures, while temperature scaling was applied to improve confidence calibration before selective decision-making.
05 - Develop an adaptive decision mechanism that	Achieved. A learned selective policy was developed to choose between crossing, non-crossing, and

can commit or request additional evidence.	OBSERVE_MORE. It achieved 88.94% committed accuracy at 87.15% coverage.
06 - Provide interpretable evidence for the resulting decision.	Achieved. Integrated Gradients was incorporated to identify supporting and opposing evidence from intent, uncertainty, observation-reliability, and motion features for individual policy decisions.
07 - Evaluate the proposed framework under dataset and runtime conditions.	Achieved within the defined research scope. Quantitative evaluation was performed using held-out PIE sequences, while automatic perception and end-to-end raw-video/CARLA experiments were used to investigate operational feasibility. These runtime experiments were treated as feasibility validation rather than a separate accuracy benchmark.

Table 13: Achievement of Research Objectives

Overall, the research achieved its stated objectives within the scope of a research prototype. The work progressed from controlled occlusion-aware data preparation to temporal prediction, uncertainty estimation, adaptive selective decision-making, explainability, and raw-video runtime evaluation. The remaining limitations relate mainly to perception robustness, track continuity under severe occlusion, broader real-world validation, and production level safety assessment.

8.3 Individual research and product contributions

The individual research contribution of this thesis is the design and evaluation of an occlusion-aware pedestrian crossing-intention framework for situations in which visual evidence is incomplete. The component combines temporal pedestrian representation, probabilistic observation reliability, Transformer-based intention prediction, calibrated uncertainty estimation, selective decision-making, and explanation within a single research pipeline.

A key scientific contribution is the explicit treatment of observation quality. Instead of assuming that every pedestrian observation is equally reliable, the component incorporates probabilistic occlusion evidence through a Bayesian observation-

reliability representation. This reliability information is then considered together with appearance, spatial, motion, and temporal evidence during intention reasoning.

The individual work also contributes an adaptive decision mechanism that can commit to crossing, commit to non-crossing, or request additional evidence. This extends conventional binary prediction by allowing uncertain observations to be deferred. The use of uncertainty measures and calibrated confidence therefore supports a more cautious decision process under partial visibility.

From a product perspective, the research was translated into a modular prototype capable of operating from raw visual input. The runtime combines pedestrian detection, tracking, occlusion estimation, reliability modelling, temporal intention prediction, uncertainty analysis, selective decision-making, and Integrated Gradients-based explanation. The component also defines clear output signals that can be consumed by downstream autonomous-driving decision-support modules.

Within the wider SOCIALMIND DRIVE project, this component represents the "Understand Intent" capability. It addresses pedestrian intention under occlusion while the other group components focus on right-of-way negotiation, socially-aware reasoning, and collaborative learning. The four components remain independently implemented and evaluated, but are brought together through a common research and demonstration framework.

Overall, the individual contribution is both research-oriented and product-oriented: it introduces a reliability-aware and uncertainty-aware approach to pedestrian-intention prediction and demonstrates how that approach can be implemented as an operational research prototype for future autonomous driving integration.

8.4 Future directions

Several directions can extend the present research beyond the current prototype. First, the automatic perception stage should be strengthened, particularly pedestrian detection, occlusion estimation, and track continuity under severe or prolonged occlusion. Improving these front end modules would reduce the propagation of unreliable observations into the intention prediction pipeline.

Second, future evaluation should use larger and more diverse annotated video datasets covering different road layouts, lighting conditions, pedestrian behaviours, traffic densities, and occlusion patterns. A fully annotated end-to-end benchmark would also allow detection, tracking, occlusion estimation, intention prediction, and selective decision making to be evaluated within the same runtime setting rather than as largely separate stages.

The current framework should also be tested more extensively in real-world and simulated urban scenarios. Broader CARLA experiments and controlled real-world video evaluation would help determine how well the model generalises beyond the PIE dataset and how strongly domain shift affects perception and intention reasoning.

Another future direction is the evaluation of real-time computational performance. Latency, memory usage, GPU requirements, and inference throughput should be measured on hardware representative of an autonomous-vehicle platform. These results would help identify whether model compression, feature optimisation, or more efficient temporal processing is required for practical deployment.

The selective decision mechanism can also be extended to consider the cost of delaying a decision. In some traffic situations, requesting additional evidence may improve confidence, whereas in others a downstream planner may require an immediate conservative response. Future work could therefore investigate decision policies that explicitly balance prediction confidence, observation reliability, available time, and potential driving risk.

Finally, the component can be integrated more closely with downstream autonomous-driving modules such as trajectory prediction, motion planning, and vehicle-control systems. Such integration would allow future research to evaluate not only whether pedestrian intention is predicted correctly, but also whether reliability-aware and uncertainty-aware intention information contributes to safer and more appropriate vehicle behaviour.

Overall, future development should move the framework from a validated research prototype toward broader end-to-end evaluation, stronger perception robustness, real time implementation, and safety-oriented system integration.

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GLOSSARY

Term	Definition
Crossing Intention	The estimated likelihood that a pedestrian intends to enter or cross the vehicle's path.
Occlusion	A condition in which a pedestrian is partially or fully hidden by vehicles, roadside objects, or other scene elements.
Observation Reliability	An estimate of how trustworthy the currently observed pedestrian evidence is, represented in this research using low-, medium-, and high-reliability probabilities.
Temporal Window	A sequence of consecutive observations used jointly for prediction. This research uses a 30-frame temporal window.
Feature Representation	The numerical description of each pedestrian observation using appearance, spatial, motion, and observation-reliability information.
Transformer	A neural network architecture based on attention mechanisms, used in this research to model temporal pedestrian behaviour and estimate crossing intention.
Predictive Uncertainty	A measure of how uncertain the model is about its estimated pedestrian intention.
Monte Carlo Dropout	A technique that performs multiple stochastic model passes during inference to estimate predictive uncertainty.
Temperature Scaling	A post-training calibration method used to improve the reliability of predicted confidence values.

Selective Decision-Making	A decision approach in which the system may either commit to a prediction or defer the decision when available evidence is insufficient.
OBSERVE_MORE	The selective-policy action used to request additional temporal evidence instead of immediately committing to crossing or non-crossing.
Decision Coverage	The proportion of observations for which the selective policy makes a committed crossing or non-crossing decision.
Committed Accuracy	The prediction accuracy calculated only over observations for which the selective policy makes a committed decision.
Integrated Gradients	An explainability technique used to estimate how individual input features contribute to a model decision.
Track Continuation	The temporary maintenance of an already-established pedestrian track when the pedestrian becomes briefly unavailable because of occlusion.
End-to-End Runtime	The execution of the complete pipeline from raw visual input through perception, intention prediction, uncertainty analysis, selective decision-making, and explanation.

APPENDICES

Appendix A Research instruments or extended results

Appendix A.1 – End-to-End Runtime Demonstration

The following figure presents the end-to-end CARLA runtime demonstration of the proposed component.



Figure 3: End-to-end CARLA runtime demonstration of the occlusion-aware pedestrian intention prediction component.

Appendix B Academic integrity and AI use disclosure

Generative AI tools were used during the preparation of this thesis to support language refinement, organisation of written content, clarification of technical explanations, and document formatting. AI-assisted suggestions were reviewed and revised before being included in the final thesis.

All research design decisions, implementation work, model development, experimental procedures, evaluation results, performance metrics, figures, and conclusions presented in this thesis are based on the actual research project and its verified artefacts. Generative AI was not used to fabricate experimental data, research results, citations, or evaluation evidence.

The final responsibility for the accuracy, interpretation, and presentation of the research remains with the author.